

ESSENTIAL PENTAHO ETL



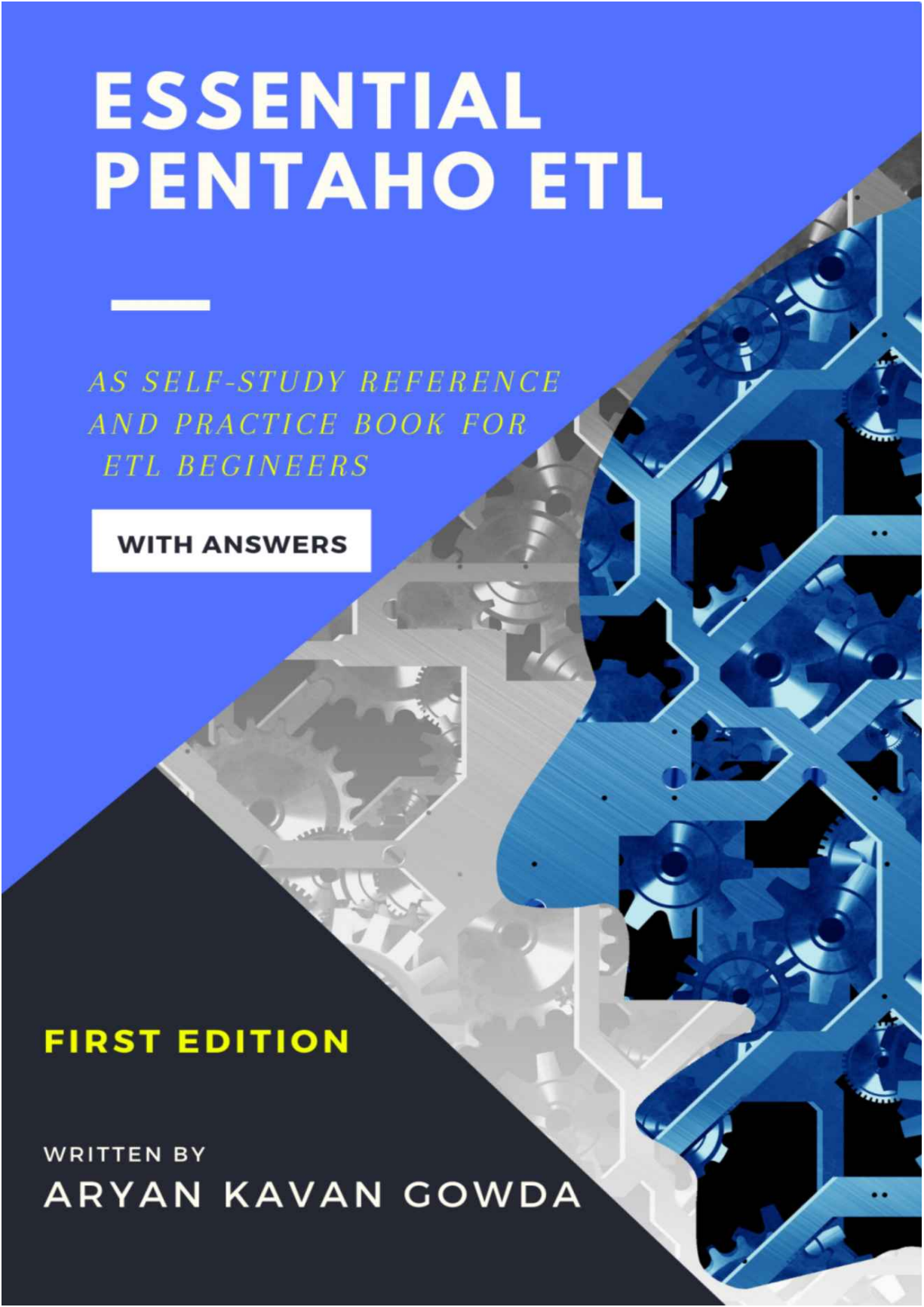
*A SELF-STUDY REFERENCE
AND PRACTICE BOOK FOR
ETL BEGINNERS*

WITH ANSWERS

FIRST EDITION

WRITTEN BY
ARYAN KAVAN GOWDA

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Essential Pentaho ETL

As self-study reference and practice book for ETL
beginners

FIRST EDITION WITH ANSWERS

— ARYAN KAVAN GOWDA

To My Parents with Love

The Valuable Resource of the 21 Century

Data Is The New Oil

-- Clive Humby

PREFACE

This book is the result of a deep interest and experience by the authors in its topic. The Essential Pentaho ETL book provides an overview of the Pentaho Data Integration. This will develop the skill to create easy ETL solutions using the GUI based designer. The Pentaho Data Integration Spoon can connect to various sources including structured, semi-structured and non-structured sources. This will also help you install the Pentaho Data Integration Spoon, customizing the Pentaho Data Integration Spoon, understanding how to deal with data, scheduling the ETL Jobs and Transformations and Monitoring the ETLs.

WHAT KIND OF BOOK IS THIS?

It is a self-study Pentaho Data Integration (PDI) ETL reference and practise book that will take you from dreaming of learning ETL and potential to unlocking your success as a ETL beginners.

So, the author of this book aims to assist you to getting acquainted with PDI ETL.

- This book describes generic PDI ETL, so it isn't dependent on a particular implementation.
- It does assume prior knowledge of BI & DWH, so it can be used more effectively.
- Topics are arranged so that the early chapters can be covered rapidly.
- Some chapters include questions and PDI exercises.
- All the chapters in this book are short enough to cover in a week or two.

WHY SHOULD YOU READ THIS BOOK?

One of the goals of this edition of Essential Pentaho ETL is to provide a book that can be used as a teach-yourself book. Here are some of the concepts that covers in the Essential Pentaho ETL, first edition, as a textbook:

- Setup PDI Spoon on your local machine.
- Setup Pentaho Data Integration server on your local machine.
- Configure the PDI.
- Customize the PDI.
- Create Jobs and Transformation.
- Create, preview, and run basic transformations containing steps and hops.
- View transformation results in the Step Metrics view and the Log view.
- Create a database connection and use it explore and interact with a data source.
- Create more transformations that involve configuring the following steps: Table input, Table output, Text file output, CSV file input, Filter, Sort rows, Row normalizer, Get Data from XML, Reading and Writing REST API data etc.
- Create Pentaho Data Integration jobs that: run transformations, use variables, load and process text files.
- Create a transformation that uses a REST API as an input and output.
- Create transformations that uses table and flat files as an input.

SYSTEM USED TO DEVELOP THIS BOOK'S ETL EXAMPLES

For the record, the Pentaho Data Integration ETL Jobs and Transformations examples in this book were developed using pdi-ce-9.1.0.0-324 with pentaho-server-ce-9.1.0.0-324 and Java 8, all running under 64-bit Windows 10.

PRE-REQUISITES

The readers should have the basic knowledge on the Windows system as the training may require them to do some basic operations. They need to have the understanding of Business Intelligence or Data Warehouse. The understanding of any of the Relational Databases and dimensional modelling will be an added advantage.

You may also need to bring a Windows/Linux Laptop, minimum with the following configuration:

1. At least 8 GB RAM on the laptop.
2. At least 40 GB of free space on Hard Drive.
3. 1.4 GHz CPU or above.
4. SUN JAVA 1.8 or above.
5. MySQL 5.x/PostgreSQL 8.x or above.

ACKNOWLEDGEMENT

Any accomplishment requires the effort of many people and this work is not different. I thank all my mentors whose patience and support were instrumental in accomplishing this task. I thank my staff for the diligent effort made this publication possible.

Most of the examples are developed and designed by the author. However, some examples are the result of a collection from various sources, such as Pentaho page and seminar participants over a few years. Regardless of the source, I wish to express my gratitude to those who may have contributed to this work, even though anonymously.

Every effort has been made to give credit where it is due for the material contained herein. If inadvertently we have omitted giving credit, the future publication will give due credit to those that are brought to the author's attention.

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Introduction

Learning ETL is an adventure of discovery, particularly because the procedure accommodates several technology paradigms, including Business Intelligence, Business Analytics, Data Warehousing, OLTP systems, OLAP system, SQL programming, Scripting language, traditional object-oriented programming and understanding of the Linux system and Windows system. Let us understand some of these technologies in detail.

Business Intelligence (BI)

Business intelligence (BI) is a set of processes, methodologies, and technologies that transform the source raw data into meaningful and useful insightful information for making business decisions. With BI tools, you can ask business related questions of your own data and receive answers in seconds. BI technologies provide analysis of historical and current data of business operations.

The common functions of business intelligence technologies are:

- Measuring, tracking and predicting sales and financial performance.
- It also makes it easy to budgeting and financial planning and forecasting.
- Predict events of business operations.
- Create and publishing reports.
- Perform Ad hoc analysis.
- Online analytical processing.
- Pull data from variety of sources for analytics.
- Data mining.
- Slice-dice of reports.
- Drilldown and drill-through of reports, etc.
- Cloud analytics.

- Strategic value driver analysis.

On-line Transaction Processing (OLTP)

Online Transaction Processing (OLTP) systems provide source data to data mart and data warehouses. A typical cash counter or POS (Point of Sale) system is an example for OLTP systems. The primary purpose of the OLTP system is to effectively function the business process/business events.

- In OLTP system, the tables are in normalized structure.
- The OLTP are volatile in nature.
- All the business operational transactions are recorded here.
- Here the insert, updates and deletes statements will be faster.
- Here the DML statements: Inserts, Update, Delete happens frequently for every business events.
- Simple queries are run against the OLTP system.
- Most of the time the OLTP system contains only current data.
- The data of OLTP shows an ongoing business process.
- OLTP design is based on the Entity Relationship.
- OLTP systems are application-oriented databases.
- The OLTP system data is archived frequently hence contains low volume of data.

On-line Analytical Processing (OLAP)

Online Analytic Processing (OLAP) systems uses the data from OLTP to answer business questions. The primary purpose of the OLTP system is to help with decision support, planning, and problem solving.

- In OLAP system, the tables are in de-normalized structure.
- The OLAP are non-volatile.

- Since it is denormalized in nature, the numbers of joins between tables are less and querying these tables are faster compared to OLTP.
- Complex business logical queries are run against the OLAP system to offer business analytical solution.
- OLAP system contains both the history data and current data for historical reporting and real-time analytical purpose.
- Online Transaction Processing is used for data analysis of the data by executives and managers.
- The OLAP let out the multi-dimensional view of all types of business activities.
- The OLAP design is based on the Star scheme, Snow flake schema.
- OLAP systems are subject-oriented databases / data marts.
- OLAP system contains larger dataset goes up to TB's of historical and aggregated data.

For example: In a Retail system, customer purchase an item and make a payment through an ATM card or any other way at a POS system. At the point of sale, the merchant counts the item and calculates the amount to prepare for an invoice for the customer. The item purchased, product quantity, shop location, POS number, datetime of transaction, customer information, production information etc are operational data elements.

OLTP System deals with operational data. Operational data are those data involved in the operation of a particular system. These operational data are then used as a source for OLAP systems for making business decision like;

1. Which is the highest selling product?
2. Who are our loyal customers?
3. Which store is selling more number products?
4. Which product category has more demand by store?

And many such complex business queries.

Data Analysis

The purpose of data analysis is to make business decisions by extracting the data from variety of sources, cleansing and transforming. The actions and decisions from the knowledge of the business data can be important and act as a foundation of success.

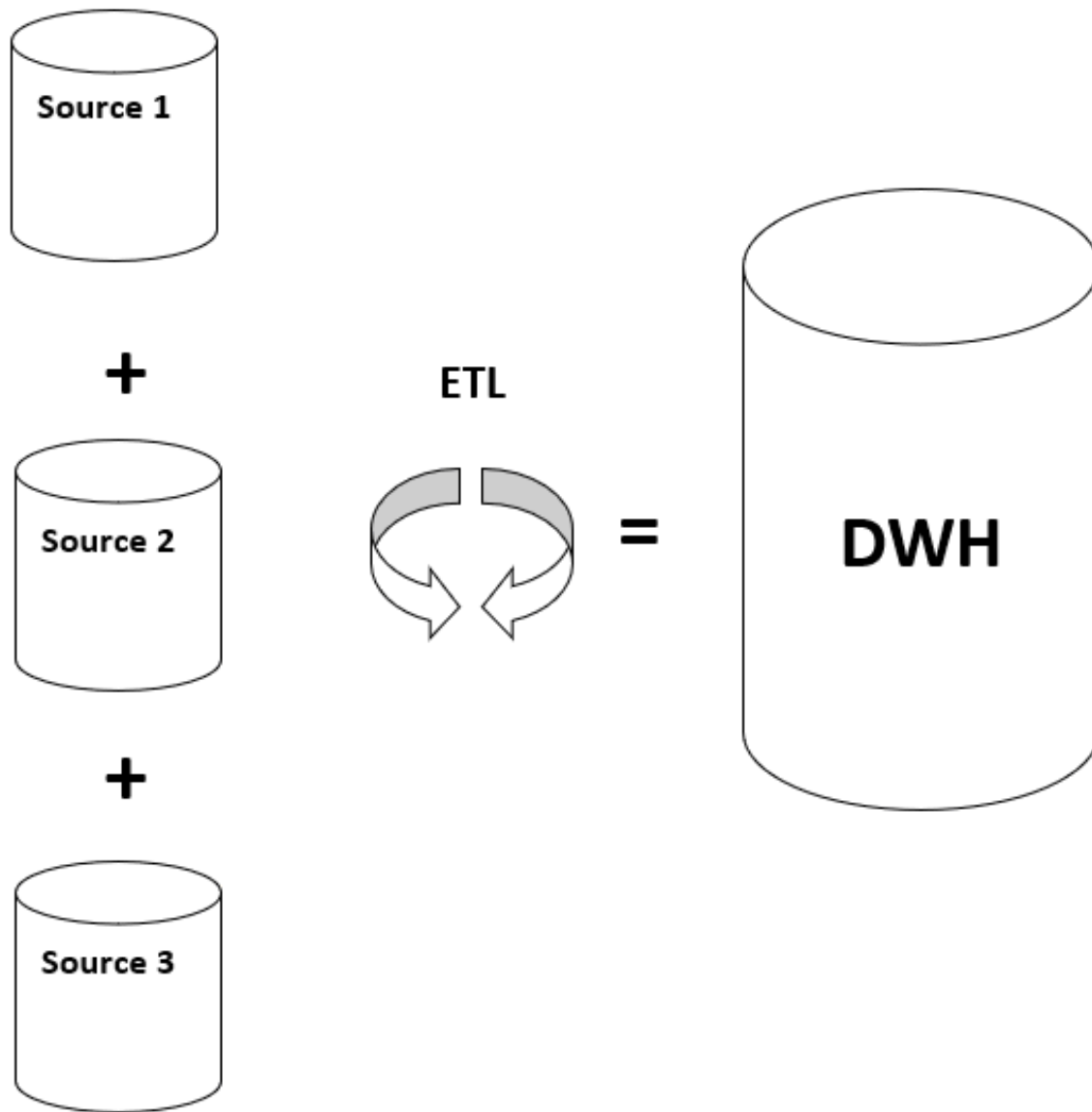
Data mining

Data mining is a process of extracting the data from large data sets for discovering the hidden patterns and developing predictive models.

Data Warehousing

A Data Warehouse is a Business Intelligence solution repository of an organization's electronically stored data. Data warehouse is an enterprise database designed to facilitate reporting and data analysis. Data warehouse technologies offers historical, current and predictive reports of business operations by analysing the present and historical business data.

Schematic View of a data warehouse



Data Warehouse is collections of data from more than 1 source. Before the data is loaded into Data warehouse or Data marts, incoming data is processed using ETL technology.

Fact & Dimension Tables

Fact & Dimension tables are main the components of a data warehouse. Facts form a part of a table called Fact table. Dimensions form a part of Dimension table. Fact table contains facts/metrics/measurements and foreign key to dimension table. Generally, facts are numbers which can be

additive, semi-additive or non-additive facts. Fact table is defined by their grain. Since the fact table contains the business process metrics, it is recommended to normalize it to reduce the data redundancy. The information which you don't know about the business are captured in fact table and the information's that you know about your business are captured in dimension table. Generally, facts are used along with their respective dimension for making business decisions. Dimensions give structure to the facts. Dimension table contains the descriptive hierarchical attributes of an entity. These attributes are used as a report filters for data analysis. The dimension tables are de-normalized in nature. Usually, one or more master table information is stored in a dimension table in a de-normalized format.

The design of a data warehouse depends on identifying facts, dimensions and granularity of the data. Generally, the relation between the facts & dimensions tables is many to one. The facts & Dimension tables are connected by a key called surrogate keys. Each fact table would have a column surrogate key that would have a corresponding key in the dimension tables. The Surrogate keys are used since they are easier to use and is an independent key of source table's primary key.

Data Integration (DI) / ETL Extract, Transform & Load

Data integration provides a mechanism to integrate the data from different variety of sources into a single database repository. It is a process of combining a large and heterogeneous collection of data sources into a data warehouse system for business analytics.

Data is pulled over from disparate sources into a data warehouse system through the process of ETL - Extraction, Transformation and Loading. This data warehouse system provides a uniform interface for querying the data

for data analysis and reporting solution. ETL layer helps to map the data from the source to target to provide a uniform data warehouse.

Pentaho Data Integration (PDI)

The Pentaho Data Integration (PDI) or Kettle or Spoon is Batch data integration and data processing tool. PDI is a GUI based ETL tool for;

- Connecting and Syncing multiple data sources.
- Processing data retrieved from multiple sources and push to multiple destinations.
- Manipulation and transforming data.
- Loading data to and from RDBMS.
- Populate DataMart and Data warehouses.

Alternatives for PDI

- Writing custom Java code.
- Writing complex Stored Procedures, Triggers, Views, Materialized views etc.
- Writing custom Scripts.
 - Perl, python, shell, java scripts etc.
- Commercial ETL tools
 - Oracle Warehouse Builder
 - Talend
 - DataStage
 - Informatica
 - SQL Server Integration services
- Open source ETL tools:
 - Talend
 - KETL
 - Clover ETL

Why Pentaho PDI ETL is better

Here is the list of benefits of Pentaho PDI Kettle over other ETL tools:

- Pentaho PDI Kettle is cost effective.
- Pentaho PDI provides graphical user interface for ETL process.
 - No connection pooling to worry about
 - No DDL to write
- Well tested full suite of components.
- Data analysis tools
 - Design steps
 - Monitoring
 - Scheduling
 - Preview
 - Data profiling with data cleaner (add on)
- Free Community edition
 - Two editions
 - GPLv2
- PDI offers high performance and throughout
 - Using table bulk loaders
- PDI leverage java
 - JVM tuning skills
 - java libraries and logic (in jars)

PDI/Kettle Data sources

- Databases
- Files
- REST APIs & HTTP Client
- XML
- Big data steps: Hadoop/HBase

- JSON
- Excel
- RSS
- Google Analytics
- And many more

Who uses PDI/Kettle?

- Countries Reached
 - 26%
- Number of downloads: 50K downloads of PDI 9.1
 - Since Sep 2020
- Different Operating Systems download
 - Windows: 41K
 - Mac: 5K
 - Linux: 3K
 - Unknown: 1K
 - Android: 99
- Kettle 10+ years old
- Open source, at version 4.4
 - GPLv2 license
 - EE edition available
 - CE edition available
- BI suite
 - C-Tools Dashboards
 - PRPT reporting
 - Cubes & analytics
 - Machine Learning

PDI/Kettle tools and files

Pentaho offers many more tools as part of the PDI client. Let us understand most common tools and files.

- Spoon – GUI tool for ETL design and development
- Kitchen – Command line interface for executing Pentaho Data Integration Jobs
- Pan – Command line interface for executing Pentaho Data Integration Transformation
- Carte Dynamic clustering tools
- Import/Export – Command line interface for executing
- Version control
- Error handling
 - Email
 - Log files
- Jobs – ETL Workflow
- Transformations – ETL Dataflow
- Slave servers – ETL slave servers

Action Plan

Understand the Business Intelligence, Data Warehouse concepts in more detail.

Chapter 1: Getting Started with PDI

In this chapter you'll learn about the following:

- Understand the system configuration for PDI server.
- Understand the system configuration for PDI client.
- Understand the components references.
- How to install Pentaho Data Integration (PDI) server.
- How to install Pentaho Data Integration (PDI)/Spoon/Kettle client.
- Install database drivers.

Welcome to PDI World! This is an exciting technology. While the former section of this book teaches you to setup and configure the PDI client and server, the latter section of this book teaches you to use the PDI Spoon, going from the transformation basics to the glory of Jobs and Transformations scheduling and monitoring. It makes the technology very powerful, but it also means there's a lot to learn.

Setting up Pentaho ETL Working Environment

PDI Server

The Pentaho Data Integration Server is hardware-independent and runs on server-class computers. The minimum hardware and operating systems specifications is given below:

Hardware—64 bit	Operating System—64 bit
• Processor	• Microsoft Windows

- Intel EM64T or AMD64 Dual-Core - RAM 8 GB with 4 GB dedicated to Pentaho DI Server - Disk Space 20 GB free after installation	2016 Server - CentOS 6 & 7 - Red Hat Enterprise 6 & 7 - Ubuntu Server 16.04 LTS & 18.04 LTS - SUSE Linux SLES 12
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At the time of publication, the latest server version is Pentaho 9.1. It is recommended to refer to the Pentaho official documentation for the latest updates.

PDI Client

The Pentaho data integration is also known as Kettle or Spoon or PDI client tool. It is hardware-independent and run-on client-class computers. The minimum hardware and operating systems specifications is given below:

Hardware—64 bit	Operating System—64 bit
- Processors - Apple Macintosh Dual-Core - Intel EM64T or AMD64 Dual-Core - RAM 2 GB RAM for most of the design tools, PDIPDI requires 2 GB dedicated	- Microsoft Windows 8 & 10 - Ubuntu Desktop 16.04 LTS & 18.04 LTS - macOS 10.13, 10.14, & 10.15

• Disk Space 2 GB free after installation • Minimum Screen Size 1280 x 960	
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At the time of publication, the latest client version is Pentaho 9.1. It is recommended to refer to the Pentaho official documentation for the latest updates.

Java Virtual Machine

Both the Pentaho server and client software's requires Java Runtime Environment (JRE). The following are the list of both certified and supported run time environments as per Pentaho.

Pentaho Software	Certified	Supported
Pentaho Server Pentaho ETL Client	• Oracle Java 8 • Oracle OpenJDK 8	• AdoptOpenJDK • Zulu from Azul Systems

- For each Pentaho release, certified list indicates the JRE is tested and validated by Pentaho.
- For non-certified JREs, support is available by Pentaho.

Solution Database Repository

Pentaho application software stores schedules, solution data processing, operations and other artifacts in the database repositories. The list of databases repositories supported by Pentaho is shown below:

- PostgreSQL 9.6* & 10.0

- MySQL 5.7 & 8.0 (SQL 92)
- Oracle 11.2 & 12.1 & 12.2 (SQL 92)
- MS SQL Server 2016, 2017

* The default installed solution database.

Web Browsers

In order to access the Pentaho User Console & Pentaho Data Integration KETTLE status pages, Pentaho supports most of the Web browsers that are publicly available. Here is list of major Web browsers supported by Pentaho:

- Apple Safari 10 & 11 (Note: On macOS only)
- Google Chrome 64 & 65
- Microsoft Internet Explorer 11 (Note: Required on Windows for data inspection and other features.)
- Mozilla Firefox 58 & 59

*Linux requires libwebkitgtk-1.0

Setting up Pentaho PDI Server

We will setup the community edition of Pentaho 9.1 server to understand the details of Pentaho server software's. The examples in this book follow the community edition interface.

Installation of Pentaho Data Integration Server

Let's understand the details about installing Pentaho Data Integration community edition.

To install the community edition of Pentaho, you should follow the steps below:

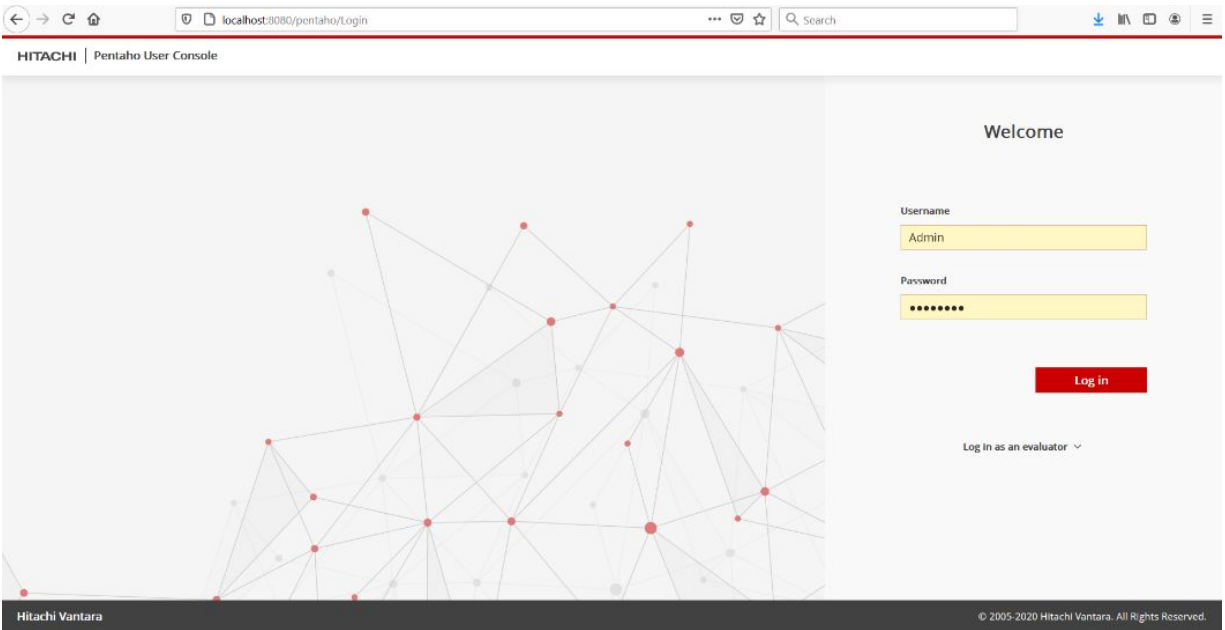
- The latest stable version (9.1) is available from the Source Forge website (<http://sourceforge.net/projects/pentaho>).

- Go to Source forge site, and navigate to *Project-> Pentaho - > Files -> Pentaho 9.1 -> server* path.
- Download the **pentaho-server-9.1.0.0-324.zip**.
- Unzip the downloaded **pentaho-server-9.1.0.0-324.zip** file.
- Windows users can copy the folder **pentaho-server** to the folder C:\Pentaho91.
- Mac and Linux users (including Oracle Virtual Box) may move the file to any folder.

Start the PDI Server

Follow the steps below to start the Pentaho server on your machine.

- Navigate to *.../pentaho-server-9.1.0.0-324/pentaho-server* path and run start-pentaho.bat file by right clicking and select “*Run as Administrator*” option.
- For Mac and Linux users, run the Spoon.sh from terminal (./spoon.sh).
- A pop-up window will appear after you run the start-pentaho.bat file that initialize the tomcat server.
- After a few minutes you should be able to see server startup message.
- That’s it! The Pentaho server is started.
- Go to <http://localhost:8080> URL to access the Pentaho User Console (PUC) page.



[Pentaho User Console Login Page]

- Enter the following credentials to login.
Username: admin
Password: password
- Alternatively, you can click on *Log in as Evaluator* options and select the user to login as.

Stop the PDI Server

To stop the Pentaho server, navigate to `../pentaho-server-9.1.0.0-324/pentaho-server` path and double click on **stop-pentaho.bat** file.

Setting up Pentaho PDI client

We will setup the community edition of Pentaho 9.1 client tool PDI Spoon as the examples in this book follow the community edition interface.

Installation of Pentaho Data Integration Client

In this edition of the book, you should install the latest 9.1 community edition of Pentaho Data Integration (PDI).

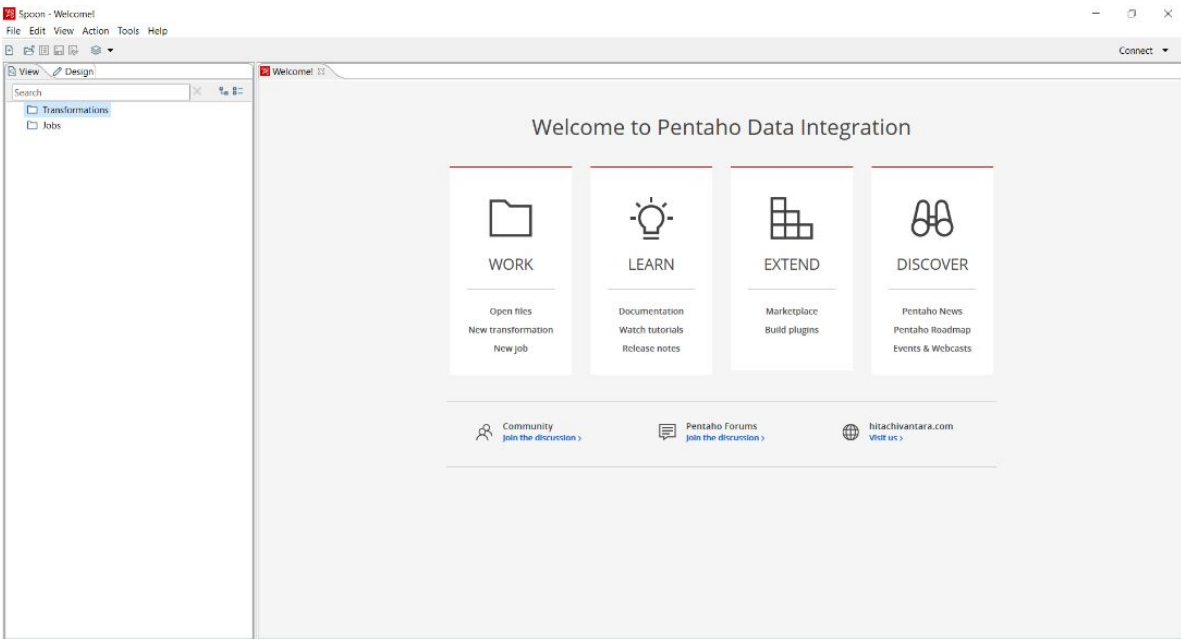
To install 9.1 PDI, you should follow the steps below:

1. The latest stable version (9.1) is available from the Source Forge website (<http://sourceforge.net/projects/pentaho>).
2. Go to Source forge site, and navigate to *Project-> Pentaho - > Files -> Pentaho 9.1 -> client-tools* path.
3. Download the **pdi-ce-9.1.0.0-324.zip**.
4. Unzip the downloaded **pdi-ce-9.1.0.0-324.zip** file.
5. Windows users can copy the **data-integration** to the folder C:\PDI91
6. Mac and Linux users (including Oracle Virtual Box) may move the file to any folder.

Start the PDI Client tool

To launch PDI client, following the steps shown below.

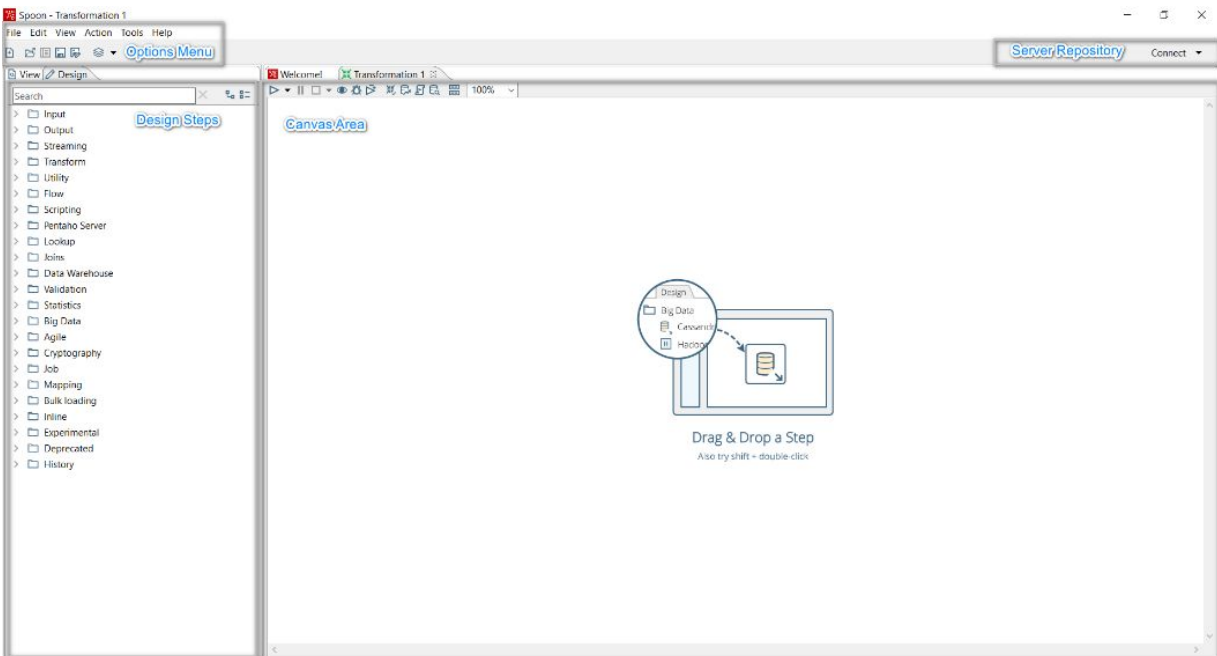
1. Run the file Spoon.bat by right clicking and select “*Run as Administrator*”.
2. For Mac and Linux users, run the Spoon.sh from terminal (./spoon.sh).
3. You will see the Welcome window with self-explanatory four options - work, learn, extend, discover as shown in the screenshot.



[Figure: Spoon Welcome Screen]

ETL Transformation Options

In order to understand the ETL Transformation option, let's create a new transformation.



[Figure: PDI Transformation screen]

The PDI transformation screen can be divided into the following sections.

1. **Canvas Area:** Here we add steps and connect them with the hops to create an ETL work flow and data flow. You can also run and preview the jobs / transformations from options above the canvas area.
2. **Options Menu:** We use Options menu to create new job/transformations, open an existing ETL files, perform save and saves options, configuring the kettle properties, setting environment variables, execute and debug ETLs, connect to the databases, connect to the repositories, explore Marketplace and etc.
3. **Design Steps:** PDI comes with a lot of preloaded plugins. Plugins are the customization that we want in our workflows.
4. **Server Repositories:** Here we connect to the PDI server where the jobs transformation is saved in the repository.

Stop the PDI Client tool

- To exit Spoon, click on **close** button in the top right corner.

Installing JDBC Drivers

Pentaho application uses the Java Database Connectivity (JDBC) to connect to databases. Most of the time while working with Spoon/PDI, you will need to connect to databases.

Here are the steps for installing the MySQL database drivers:

In order to connect to the database, you must download and install the appropriate database driver.

Steps

1. Go to <http://dev.mysql.com/downloads/connector/j>
2. Download “mysql-connector-java-5.1.44-bin.jar” zip file format.

3. Unzip the file.
4. Copy the jar file to both Pentaho server and PDI Spoon tool.

Pentaho Server

1. Stop the Pentaho server.
2. Copy the mysql-connector-java-5.1.36-bin.jar file “*.../pentaho-server/tomcat/lib*” folder.
3. Restart Pentaho server.

For the other database drivers, repeat the same steps above after downloading the respective the drivers.

Pentaho Client

1. Stop the Pentaho PDI Spoon client.
2. Copy the mysql-connector-java-5.1.36-bin.jar file to “*.../data-integration/lib*” folder.
3. Restart both Pentaho Data Integration Spoon.

For the other database drivers, repeat the same steps above after downloading the respective the drivers.

Action Plan

Setup PDI Server and PDI client (Spoon) on your development machine.

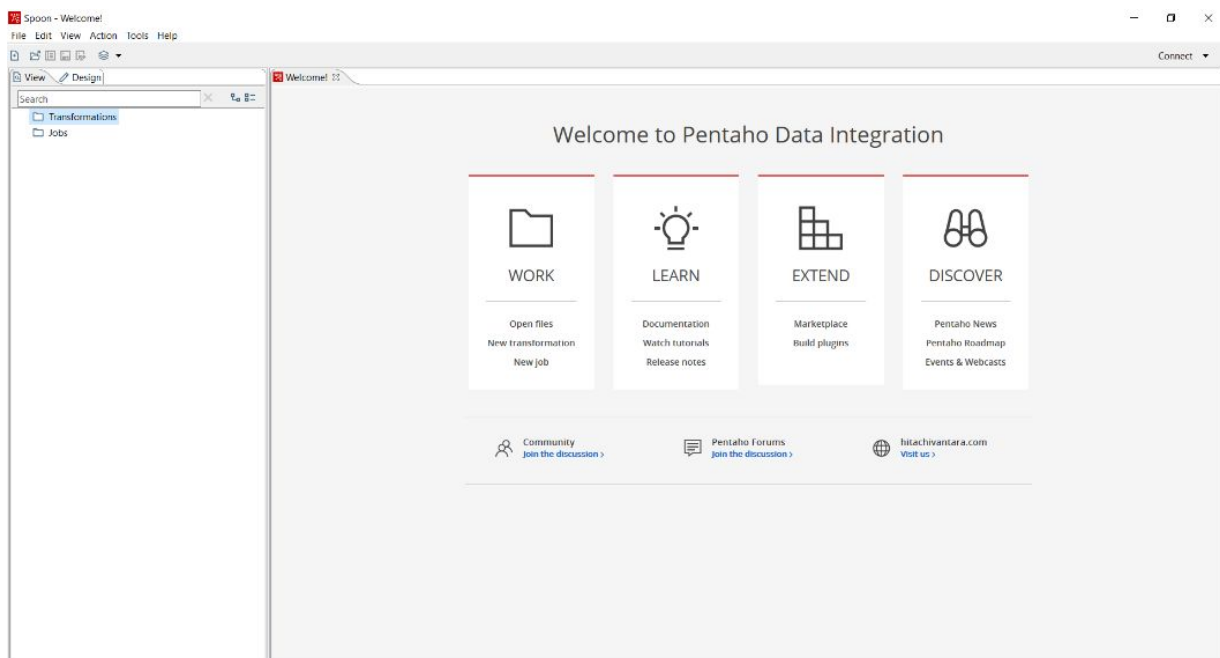
Chapter 2: Customize Your PDI

In this chapter you'll learn about the following:

- Customization of PDI Spoon client tool.
- PDI Marketplace

Kettle Options

We can customize the general kettle options and look & feel of the PDI Spoon client tool as per our interests. Let us understand this by launching the Spoon using Spoon.bat (Windows) or Spoon.sh (Linux) based on your OS.



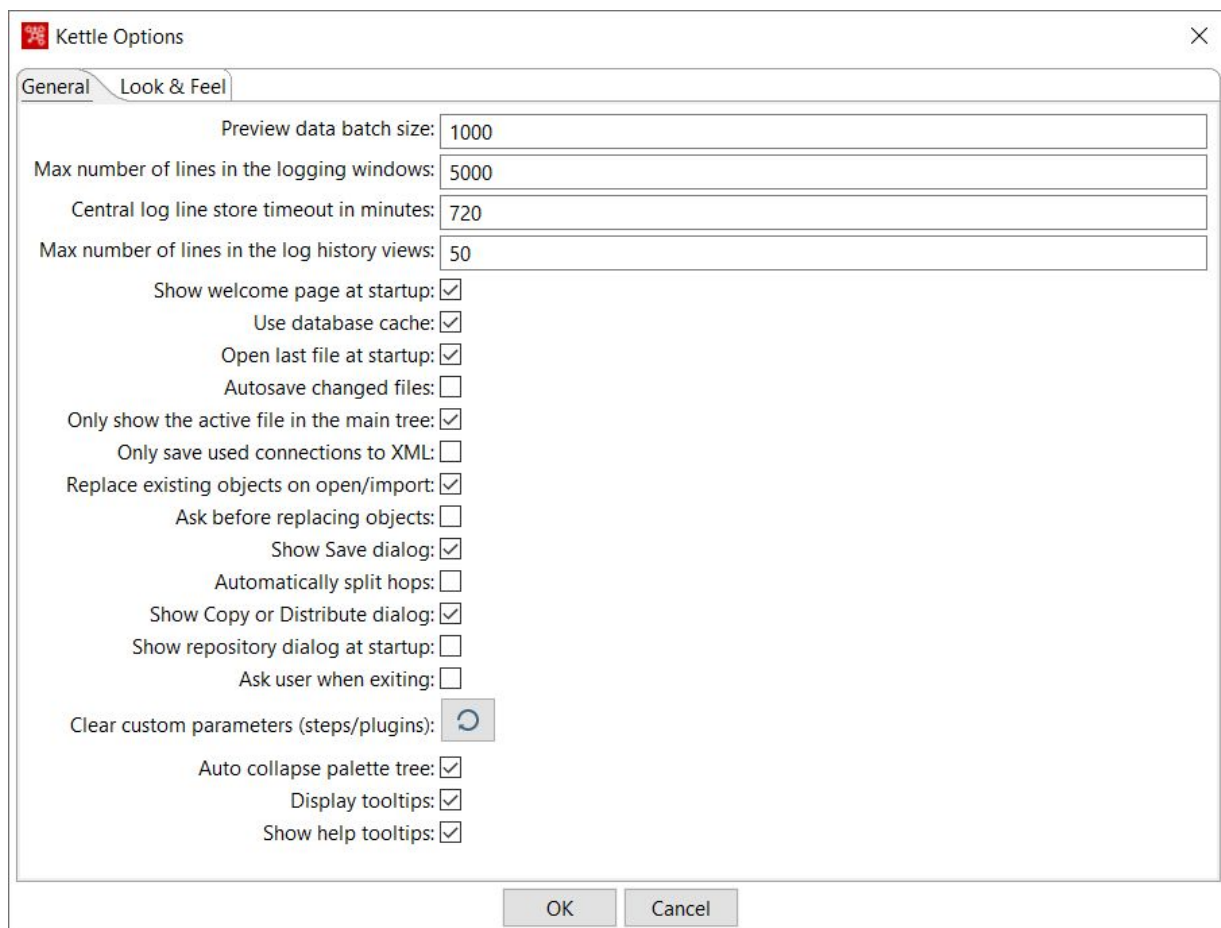
[Figure: Spoon welcome screen]

Pentaho allows us to customize the PDI Client (Spoon) tool as per our preferences. In order to change these settings, click on the **Tools** options

from navigation bar and then click on the **Options**. You will see the **Kettle Options** window as shown below.

General

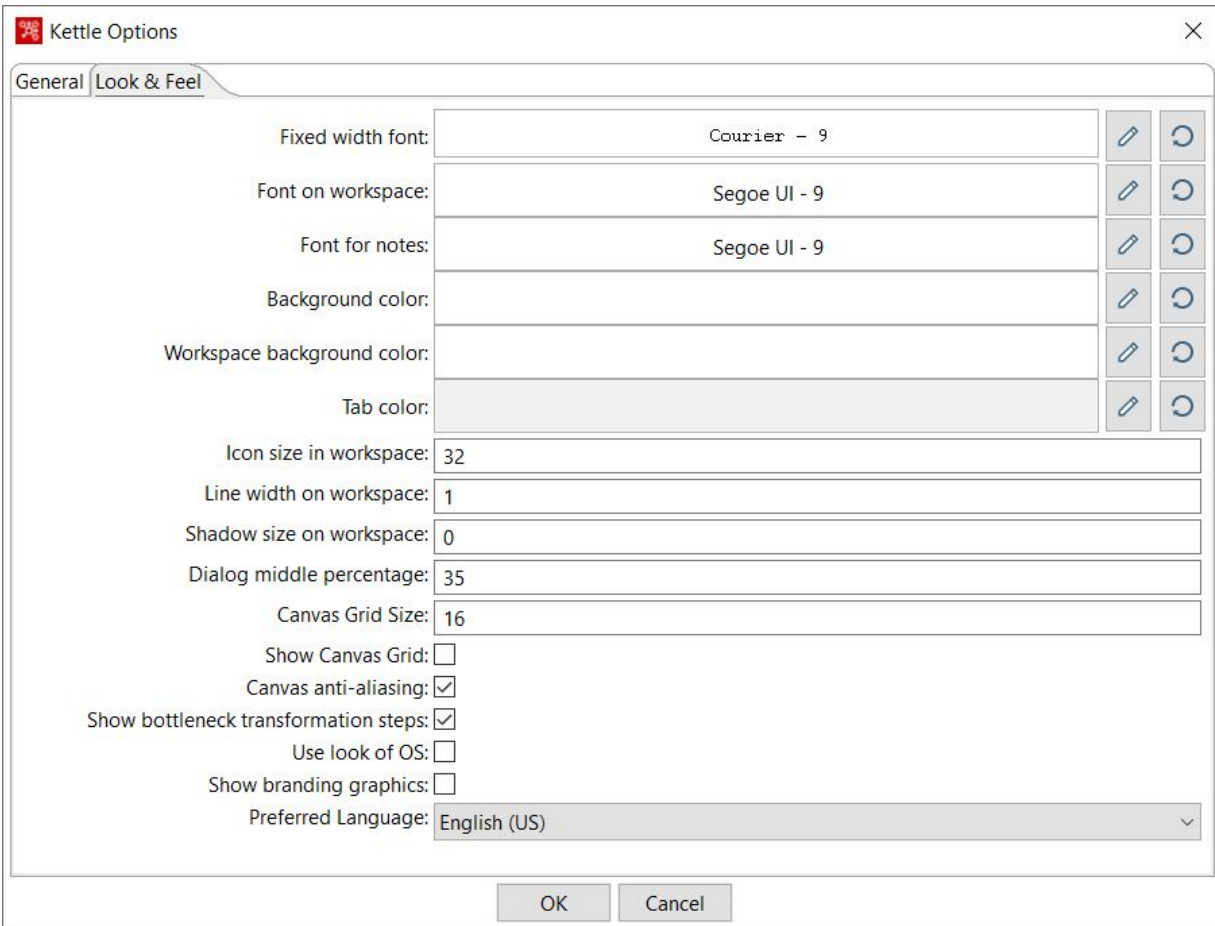
In the General tab, there are some options related to the transformation and job settings, preview, logging etc. Update the settings as per your requirements.



[Figure: Kettle Options – General tab]

Look & Feel

In the General tab, you can customize the look and feel of the PDI Spoon tool. You can customize the Background colour, font size, font colour, tab colour, Icon size many other options. Make the changes as per your needs.



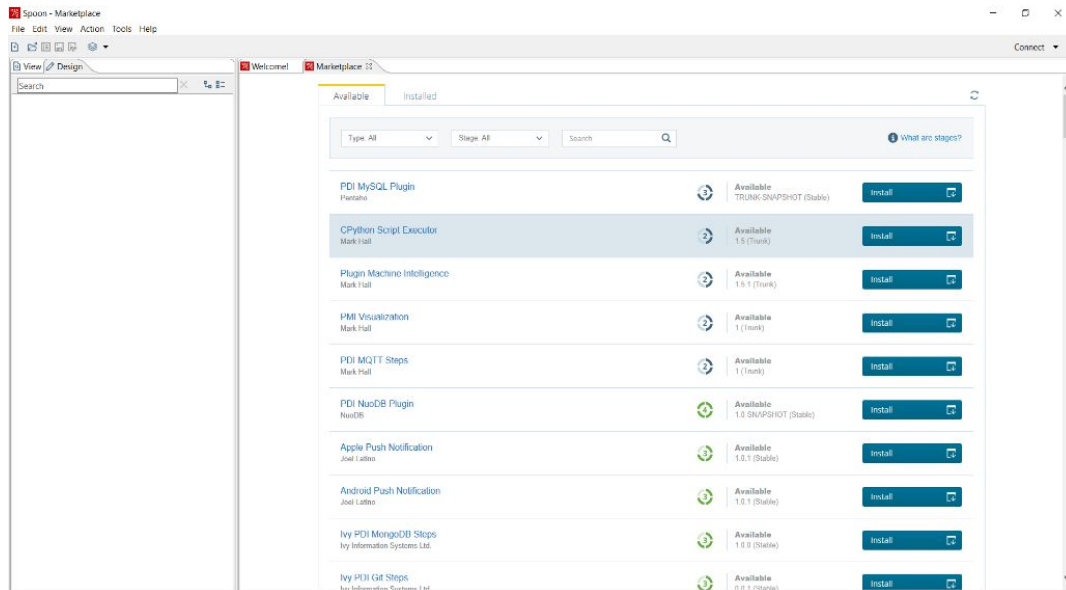
[Figure: Kettle Options – Look & Feel tab]

PDI Marketplace

Pentaho offers flexibility to install the third-party PDI plugins that includes Bigdata, Business Intelligence, connectivity plugins, data mining, IoT, statistics and many other plugins. These plugins are categorized into 2 parts- Customer and Community. We are free to install these marketplace plugins.

Installation Steps:

1. Launch the Spoon by running `Spoon.bat` on windows. For Mac and Linux users, run the `Spoon.sh` from terminal (`./spoon.sh`).
2. Go to **Tools** and select **Marketplace** option.



3. Select the relevant plugins from the list and click on Install button.
4. Restart the PDI application for the changes to take effect.

All the plugins installed plugins are stored in `.../data-integration/plugins` folder.

Disclaimer: Please make sure it a stable release before using it on production.

Action Plan

Customize your PDI Spoon client as per your development preference.

Chapter 3: Configure Your PDI

In this chapter you'll learn about the following:

- Configure PDI Spoon to connect the PDI server
- Configure Property files

In order to work the PDI Spoon in a development environment you need to configure the repository, kettle property files and logs.

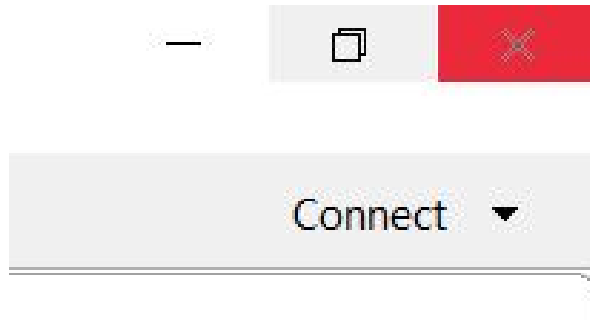
Configure the PDI Repository

PDI server repository is a central place which contains all the ETL's jobs & transformations. All the developers connect to this central repository for the development.

Here we will understand how to configure the PDI client to access the PDI server repository.

Steps:

1. Start the Pentaho server by running `start-pentaho.bat` from the installed directory.
2. Launch the Spoon by running `Spoon.bat` on windows. For Mac and Linux users, run the `Spoon.sh` from terminal (`./spoon.sh`).
3. PDI welcome screen appears.
4. In the top-right corner, click the Connect button as shown in the screenshot.



5. The Pentaho Repository Manager dialog box appears. Click **Add** or **Get Started** button.
6. Fill the connection details as shown in the screen shot.

A screenshot of the 'Repository Manager' dialog box. The title bar says 'Repository Manager' with a red Pentaho logo on the left and a close button on the right. The main content area is titled 'Connection Details'. It contains three text input fields: 'Display Name' with the value 'ETLServer90', 'URL' with the value 'http://localhost:8080/pentaho', and 'Description' with the value 'Pentaho repository | http://localhost:8080/pentaho'. Below these fields is a checkbox labeled 'Launch connection on startup' which is currently unchecked. At the bottom left is a 'Help' button with a question mark icon. At the bottom right are 'Back' and 'Finish' buttons.

- **Display Name:** Enter the Display Name property.
- **URL:** Modify the URL associated with your repository, if necessary.

- **Description:** Enter the Repository description
- **Launch connection on startup:** check this option if you want connect to the repository on startup (Optional).

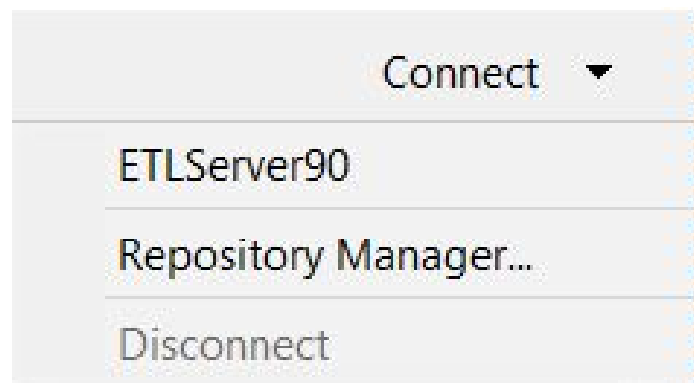
7. Click Finish to test the connection of your repository.

Connecting to the PDI Repository

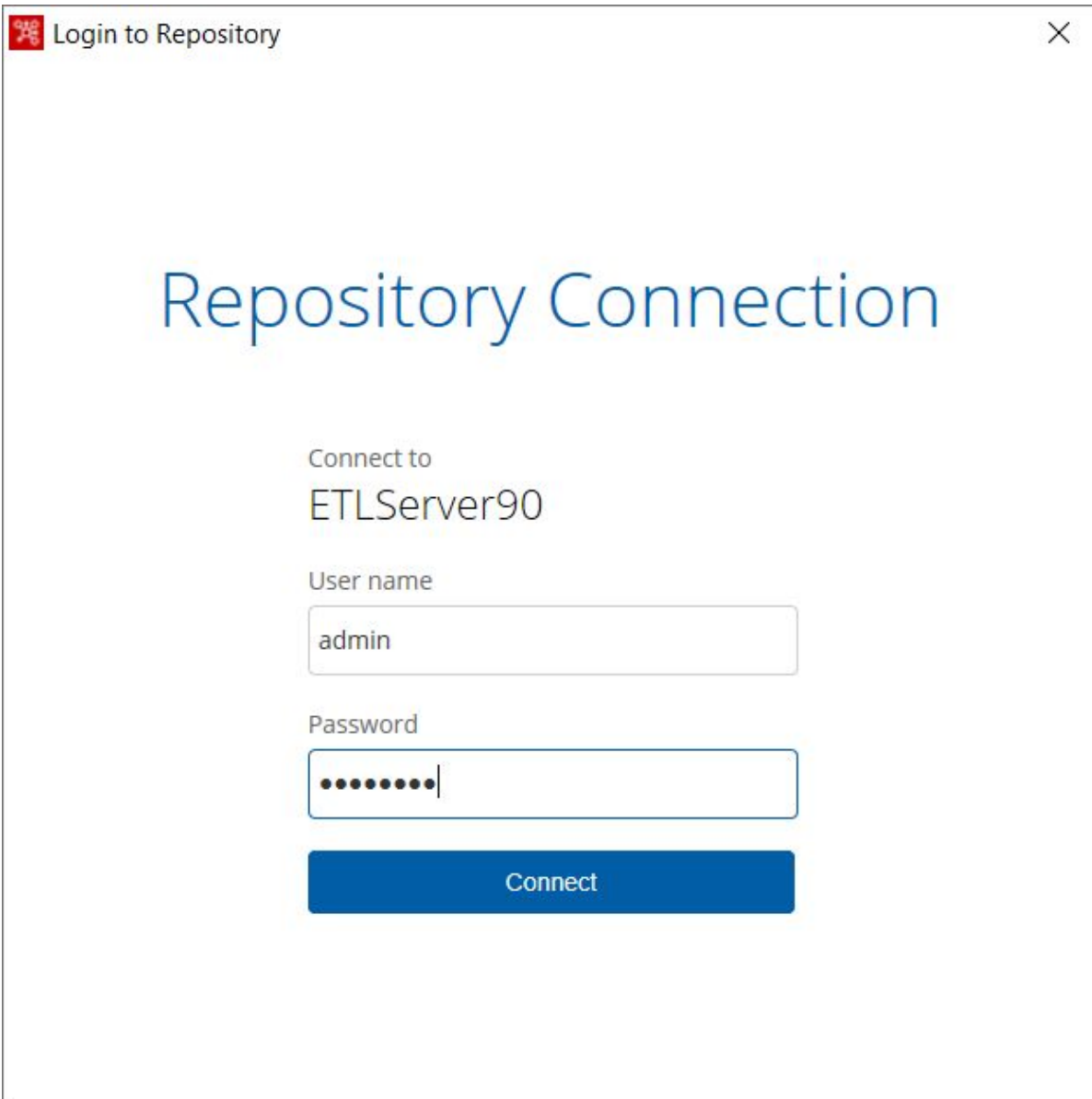
Follow the steps below to connect to the PDI repository from Spoon client.

Steps:

1. Launch the Spoon by running `Spoon.bat` on windows. For Mac and Linux users, run the `Spoon.sh` from terminal (`./spoon.sh`).
2. PDI welcome screen appears.
3. In the top-right corner, click the `Connect` button as shown in the screenshot.



4. Select the respective connection from the Connect drop-down menu in the top-right corner of the screen.
5. Enter the PDI server connection details as shown in the screenshot.



Login to Repository

Repository Connection

Connect to
ETLServer90

User name
admin

Password
.....

Connect

- Username: Pentaho Server username
- Password: Pentaho Server password

6. Click on **Connect** button.

Disconnecting the PDI Repository

In the top-right corner of the PDI Spoon screen, click on the Connection dropdown and choose **Disconnect** option to disconnect from PDI server

repository.

Configure KETTLE Home

The default KETTLE Home directory is the user's home directory. This directory may change depending on the user who is logged on. Hence variable values used in the PDI jobs and transformations may change from user to user. To overcome such issues, we set the KETTLE_HOME variable in both PDI client and PDI server. By doing so, the jobs and transformation will give the desired results irrespective of the user who has logged in.

The Kettle folder contains the following files.

1. **kettle.properties** - The properties file for variables.
2. **shared.xml** - The shared objects file such as DB connection.
3. **db.cache** - The database cache.
4. **repositories.xml** - The local repositories file.
5. **.spoonrc** - User interface settings, last opened transformation/job.
6. **.languageChoice** - User language.

Follow the steps below to set the KETTLE_HOME variable.

1. Open the Pentaho server startup script `start-pentaho.bat`.
2. Append “-DKETTLE_HOME=/path/to/you/kettle_home” option at the end of the CATALINA_OPTS variable.

```
Ex: set CATALINA_OPTS=-Xms2048m -  
Xmx6144m -XX:MaxPermSize=256m -  
Dsun.rmi.dgc.client.gcInterval=3600000 -  
Dsun.rmi.dgc.server.gcInterval=3600000 -  
Dfile.encoding=utf8 -  
DDI_HOME=%DI_HOME% -  
DKETTLE_HOME=C:\Users\Admin\.kettle
```

(OR) Try the following setting to set the KETTLE home system wide.

1. Go to, **System -> System Properties -> Environment variables**
2. Create a new System variable and add the following details:
 - **Variable Name:** KETTLE_HOME
 - **Variable Value:** C:\Users\Admin\.kettle
3. Click Ok button to add the new environment variable
4. Stop the Pentaho server
5. Clear the contents of the following folder:
 - /tomcat/logs/
 - /tomcat/temp/
 - /tomcat/work/
 - /pentaho-server/logs/
 - /pentaho-solutions/system/karaf/caches/
 - /pentaho-solutions/system/jackrabbit/repository/
6. Start the Pentaho server

Action Plan

- Configure your PDI client (Spoon) to PDI server.
- Configure the KETTLE_HOME directory.

Chapter 4: Dealing with Data

In this chapter you'll learn about the following:

- Reading files using PDI Spoon
- Reading tables using PDI Spoon
- Reading REST API data using PDI Spoon

The PDI ETL tool has several steps for dealing with different formats of data. The Pentaho PDI Spoon has several steps grouped by the category i.e., input, output, transformation, streaming, statistics, big data, scripting, data warehouse, bulk loading steps that allow you to read, write and transform all the structured, semi-structured and non-structured data. In this chapter, you will learn not only the basics for reading and writing data, but also all the how-to's for dealing with them.

Working with Files and Directories

When working with ETL tools, reading and writing the flat files is constant and habitual. With PDI, reading and writing files is an effortless task. You pick the design step, configure it quickly, and you are done. Let us understand this concept by reading a sample plain file.

Reading a TXT file

In this example, we have to read a simple `source-customer-info.txt` file with a list of customer information.

```
ID;Name;Age;City  
1101;Jack;20;Bangalore  
1102;Ilisa;25;Hassan
```

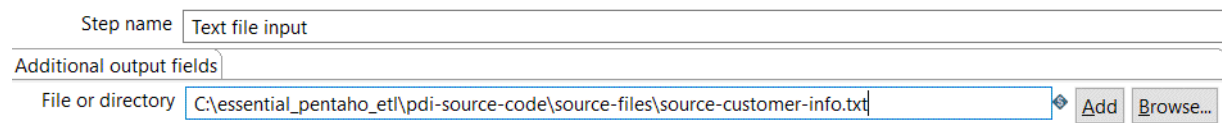
1103;Kumar;23;Mysore

1104;Kavan;26; Mandya

ETL Flow:

Carry out the following steps:

1. Launch the PDI Spoon using Spoon.bat on Windows or Spoon.sh on Linux/Mac.
2. Create a new transformation and save the file as " read-txt-data.ktr ".
3. In order to read text files, in this case, the source-customer-info.txt file we use the **Text file input** step.
4. Drag and drop a **Text file input** step (Input category) from the **Design tab** to the canvas area.
5. Edit the step by double clicking on it. Now, in the **File or directory** textbox, you have to select the name of the file (source-customer-info.txt) with its complete path. After browsing the file, the textbox will be populated with the complete path of the file.

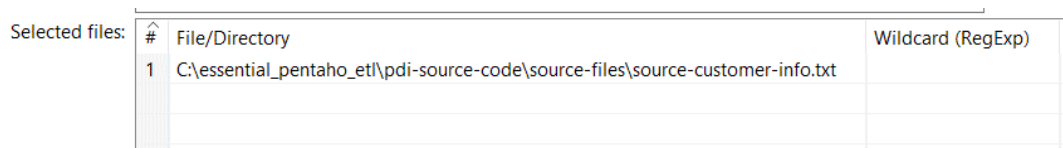


Step name

Additional output fields

File or directory

6. Click on the **Add** button. The complete text will be moved from the **File or directory** textbox to the **Selected files** grid.



#	File/Directory	Wildcard (RegExp)
1	C:\essential_pentaho_etl\pdi-source-code\source-files\source-customer-info.txt	

7. Select the **Content** tab and fill in the required fields, as shown in the following screenshot:

Filetype	CSV
Separator	;
Enclosure	"
Escape	
Header	☒ Number of header lines 1

By analysing the content of the source file, we can see that the filetype is **CSV** hence we select *CSV option* as a Filetype. We set ';' to indicate the field's separator. We also need to indicate the Enclosure string, in this case, ". Finally, we see that the first line contains the header of the columns. For the PDI to recognize that header, we have to check the **Header** checkbox and type 1 in the number of header lines textbox.

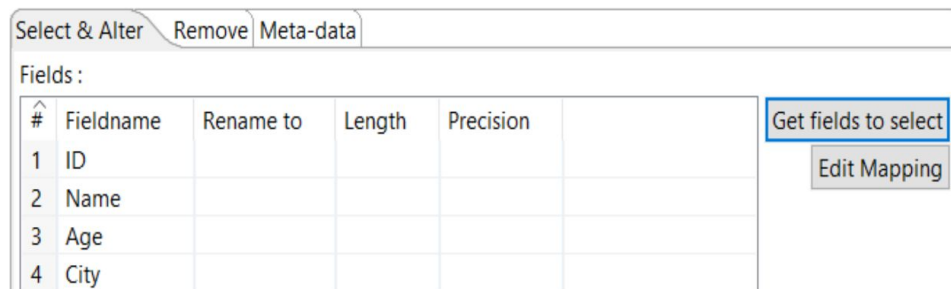
8. Select the **Fields** tab and click on the **Get Fields** button to get the definitions of the fields automatically. The grid will be populated, as shown in the following screenshot:

File	Content	Error Handling	Filters	Fields	Additional output fields								
#	Name	Type	Format	Position	Length	Precision	Currency	Decimal	Group	Null if	Default	Trim type	Repeat
1	ID	Integer	#		15	0	\$	.	,	-		both	N
2	Name	String			5		\$	.	,	-		both	N
3	Age	Integer	#		15	0	\$	.	,	-		both	N
4	City	String			9		\$	.	,	-		both	N

PDI engine takes all that information and uses it to parse the text file and fill the fields correctly. The PDI engine will automatically select the field name, type, max size i.e., length,

precision. We can always go back and edit these settings as per the business needs.

9. Click on the **Preview** button and you will see some sample rows built with the data in your file.
10. In order to alter the field names, reorder, or modify the metadata information of the columns coming from the source file, PDI spoon offers an easy solution for it. For such cases, we use **Select values** step.
11. Let us modify and reorder some of the columns we read from the source file in the same transformation file.
12. Add a **Select values** step to the Canvas.
13. Edit the step and click Get fields to select button in the right side of the screen.



The screenshot shows the 'Select & Alter' dialog box in PDI. It has three tabs: 'Select & Alter' (active), 'Remove', and 'Meta-data'. Below the tabs, there is a section labeled 'Fields :'. Inside this section is a table with the following columns: '#', 'Fieldname', 'Rename to', 'Length', and 'Precision'. The table contains four rows of data:

#	Fieldname	Rename to	Length	Precision
1	ID			
2	Name			
3	Age			
4	City			

To the right of the table, there are two buttons: 'Get fields to select' (highlighted with a blue border) and 'Edit Mapping'.

14. Suppose you want to rename the field *City* to *Address* and move it to the third row, in this case, the **Select value** step allows you to rename and reorder the field. Under the Select & Alter tab, select all the fields and manipulate those according to your needs as shown in the following example:

Select & Alter Remove Meta-data					
Fields :					
#	Fieldname	Rename to	Length	Precision	
1	ID				
2	Name				
3	City	Address			
4	Age				

15. Suppose you want to modify the meta-data information of the field Address. Change the Length of the field from 9 size to 100 size. The changes are as shown in screenshot:

Select & Alter Remove Meta-data					
Fields to alter the meta-data for :					
#	Fieldname	Rename to	Type	Length	
1	Address		String	100	

You will see that there are more options to set, but the once that are shown here are by far the most used options. But there are few options that needs to be considered as per your need and that might interest you.

The PDI Spoon offers more options to achieve a task. For examples, in this case, to work with these kinds of delimited text files, you could choose the CSV file input step. However, this step has a less configuration options, but it provides better performance. With Kettle, it is possible to read more than one file at a time using a single Text File Input step.

Reading a CSV files using CSV step Input

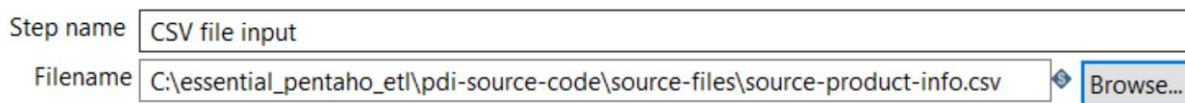
In this example, we read a CSV (Comma Separated Values) file type. We will read a simple `source-product-info.csv` file with a list of product information.

```
ID,Product,Price,Model
1101,"Eagle Light",1420,2015
1102,"Elegant Watch",2451,2018
1103,"Classic Engine",6845,2019
1104,"Blue, Moon",9462,2020
```

ETL Flow:

Carry out the following steps:

1. Launch the PDI Spoon using `Spoon.bat` on Windows or `Spoon.sh` on Linux/Mac.
2. Create a new transformation and save the transformation file as "`read-csv-data.ktr`".
3. Drop a **CSV file input** step (Input category) from the Design tab to the canvas area.
4. In the `Filename` textbox, select the directory and the files. I.e., you have to select the name of the file (`source-product-info.csv`) with its complete path.
5. After browsing the file, the textbox will be populated with the complete path of the file as shown in the screenshot:



The screenshot shows a configuration window for a 'CSV file input' step. The 'Step name' field is labeled 'CSV file input'. The 'Filename' field contains the path 'C:\essential_pentaho_etl\pdi-source-code\source-files\source-product-info.csv'. To the right of the 'Filename' field is a 'Browse...' button with a magnifying glass icon.

6. Select the **Delimiter and Enclosure textbook** fill in the required fields, as shown in the following screenshot:

Step name

Filename

Delimiter

Enclosure

NIO buffer size

Lazy conversion? ☒

Header row present? ☒

By analysing the content of the source file, we can see that the filetype is **CSV**. We set ',' to indicate the **Delimiter** separator. We also need to indicate the **Enclosure** string, in this case, ". Finally, we see that the first line contains the header of the columns. For the PDI to recognize that header, we have to check the **Header row present?** checkbox and type **1** in the number of header lines textbox. And leave the options as it is.

- Click **Get fields** button in the bottom of the pop-window to select all the source fields.

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	ID	Integer	#	15	0	\$	.	,	both
2	Product	String		14		\$	.	,	both
3	Price	Integer	#	15	0	\$	.	,	both
4	Model	Integer	#	15	0	\$	.	,	both

- Click on the **Preview** button and you will see some sample rows built with the data in your file.

Read XML data using XML Input Stream step

In this example, we read an XML file type. We will read a simple source-product-info.xml file with a list of product information.

ETL Flow:

Carry out the following steps:

- Create a transformation and save the transformation file as "read-xml-data.ktr".

2. Use an **XML Input Stream (StAX)** step to read the file source-product-info.xml.
3. In the **Filename** textbox, select the directory and the files. I.e., you have to select the name of the file (source-product-info.xml) with its complete path.
4. After browsing the file, the textbox will be populated with the complete path of the file as shown in the screenshot:

Step name

Filename

5. In the **Encoding** text box, choose the xml encoding type.
6. In the **Default Length String** text box, choose the xml element/attributes value length.
7. In the same tab, fill in the required fields, as shown in the following screenshot:

Include filename in output?	☒	Fieldname	xml_filename
Row number in output?	☒	Fieldname	xml_row_number
XML data type (numeric) in output?	☒	Fieldname	xml_data_type_numeric
XML data type (description) in output?	☒	Fieldname	xml_data_type_description
XML location line in output?	☒	Fieldname	xml_location_line
XML location column in output?	☒	Fieldname	xml_location_column
XML element ID in output?	☒	Fieldname	xml_element_id
XML parent element ID in output?	☒	Fieldname	xml_parent_element_id
XML element level in output?	☒	Fieldname	xml_element_level
XML path in output?	☒	Fieldname	xml_path
XML parent path in output?	☒	Fieldname	xml_parent_path
XML data name in output?	☒	Fieldname	xml_data_name
XML data value in output?	☒	Fieldname	xml_data_value

8. Click on the **Preview** button and you will see some sample rows built with the data in your file.

Working with Tables and Databases

All the OLTP system uses databases to store and manage transactional data such as bank transactions, purchases and sales, airport/airline operations, medical records, social data, government data and so on. Database is the storage method for **data warehouses**. Database is the target repositories used as a source for reporting in Business Intelligence solutions.

Here you will learn how to work with databases tables and views in Pentaho PDI Kettle.

PDI - SQL dialect-specific

Pentaho software generates SQL dialect when communicating with these data sources connections. Following are the certified and non-certified connection sources as per the Pentaho.

Certified

Certified indicates the SQL dialect has been tested for compatibility with Pentaho. Here is the list of certified sources:

- Amazon Redshift
- Hive
- Hive 2
- MS SQL Server (JTDS Driver)
- MS SQL Server (Microsoft Driver)
- MySQL
- Oracle
- Oracle RDB
- PostgreSQL
- Vertica

Non-certified

Non-certified indicates the SQL dialect has been tested for compatibility with Pentaho. Here is the list of Non-certified sources:

- Apache Derby

- AS/400
- InfiniDB
- Exasol 4
- Firebird SQL
- Greenplum
- H2
- Hypersonic
- IBM DB2
- Impala
- Infobright
- Informix
- Ingres
- Ingres VectorWise
- MaxDB (SAP DB)
- MonetDB
- Neoview
- Netezza
- SAP HANA
- Snowflake
- SQLite
- Teradata
- UniVerse database
- Other SQL-92 compliant*

If your data source is not in this list and if it is compatible with SQL-92, you can use a generic SQL dialect to establishing the database connection. It is recommended to refer to the Pentaho official page, for the latest and UpToDate information on this.

Connecting to a database

A database connection allows us to access a database from PDI Kettle environment. Once we define and establish a database connection, we will be able to query those database objects by executing SQL statements: run SELECT, INSERT, DELETE statements, modify rows, create and query database objects like tables, views, refresh views, call sub-routine and so on.

Most of time while working with Pentaho PDI KETTLE, you intend to work with a database, either queries, loading, looking up data, updating, and so on, the first thing you will have to do is to create a connection to that database. In order to create the database connection from your PDI Spoon, you need have access to the database from the working environment and you will need to know the connection settings. At the minimum you will need:

- **Host Name** : Domain name or IP address of the database server.
- **Port Number**: Port number of the database server
- **User Name**: Username of the database server
- **Password**: Password of the database server

Once you have these details, The PDI spoon offers different ways to connect to the database. Here is how to create the database connection.

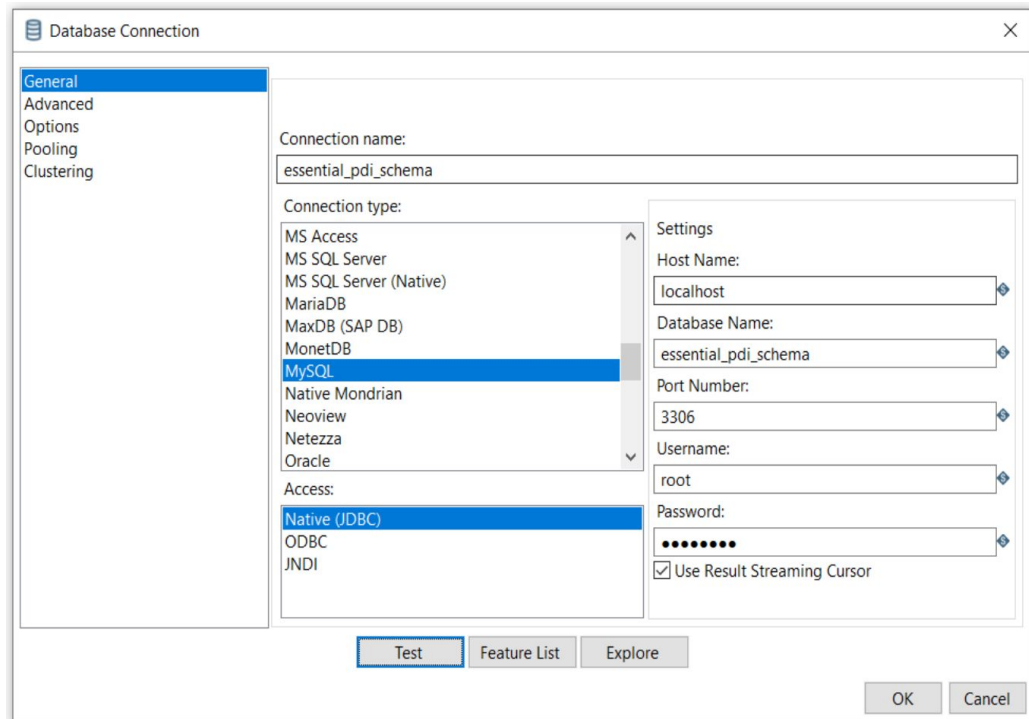
Steps:

1. Launch Spoon.bat on Windows or Spoon.sh on Linux machine.
2. Select **File -> New - Database Connection** option
(OR)
3. Create a new Transformation and select the **View** option that appears in the upper-left corner of the screen, right-click the **Database connections** option, and select **New**.
(OR)
4. Select **Tools -> Wizard -> Create Database Connection** option.

5. Try any of these 3 ways, the **Database Connection** dialog window appears similar as shown in the screenshot:

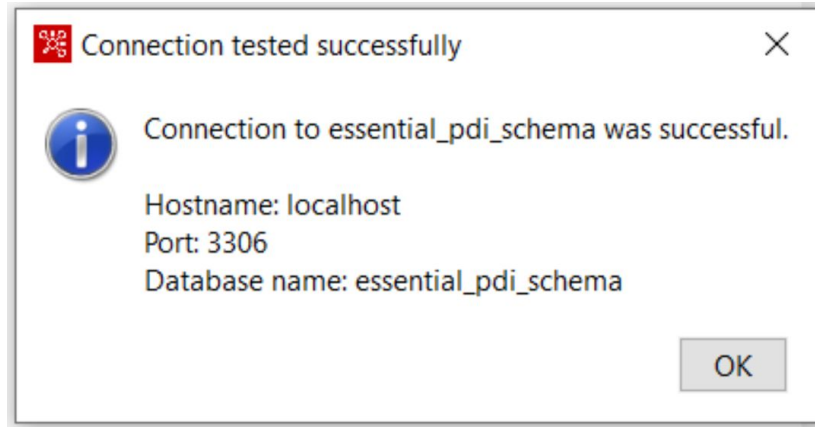
The screenshot shows the 'Database Connection' dialog window. On the left, a sidebar contains tabs: 'General' (selected), 'Advanced', 'Options', 'Pooling', and 'Clustering'. The main area is divided into two sections. The top section, labeled 'Connection name:', has a text input field. Below it, the 'Connection type:' section features a list box with the following items: 'Oracle' (selected), 'Oracle RDB', 'Palo MOLAP Server', 'Pentaho Data Services', 'PostgreSQL', 'Redshift', 'Remedy Action Request System', 'SAP ERP System', 'SQLite', 'Snowflake', and 'SparkSQL'. The bottom section, labeled 'Access:', has a list box with 'Native (JDBC)' (selected), 'ODBC', 'OCI', and 'JNDI'. To the right of these lists is a 'Settings' panel with several fields: 'Host Name:', 'Database Name:', 'Tablespace for Data', 'Tablespace for Indices', 'Port Number:' (with the value '1521' entered), and 'Username:'. At the bottom of the dialog are three buttons: 'Test', 'Feature List', and 'Explore'. In the bottom right corner are 'OK' and 'Cancel' buttons.

6. Click on the **General** Tab and fill in the details as shown below:

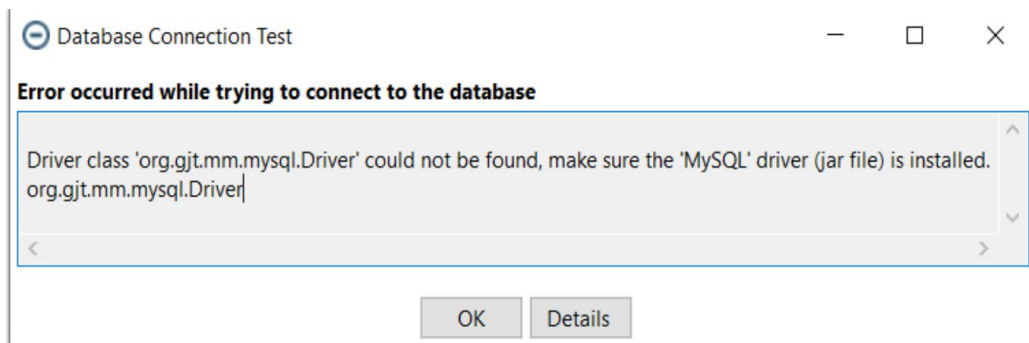


- **Connection name:** Name of the DB connection
- **Connection type:** Database Type
- **Access:** Access Method (JDBC, ODBC, JNDI)
- **Host Name or IP:** Database server name or IP address
- **Database name:** Database name you are trying to connect
- **Port Number:** Database port number
- **User Name:** Database username
- **Password:** Database password

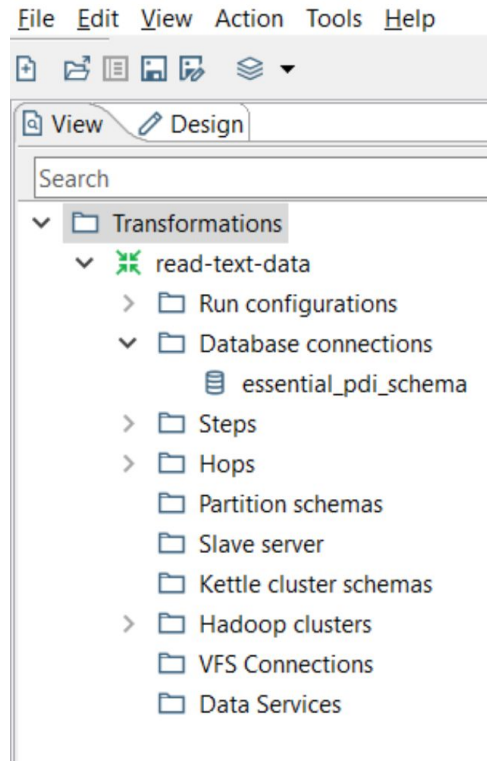
7. Enter the database connection name in the **Connection name** textbox.
8. Under **Connection Type**, select the database engine of your source/target DBMS.
9. Fill all the database connection details in the **Settings** options.
10. Press the **Test** button. A message should appear informing you that the connection to your database is **OK**.



If you face connection issues related to the database driver as shown in the screenshot:



Copy the database driver to the {...\data-integration\lib} path and restart the PDI Spoon client. Retest the connection again. All the connections are listed under the **View -> Transformation -> Database connections** section.



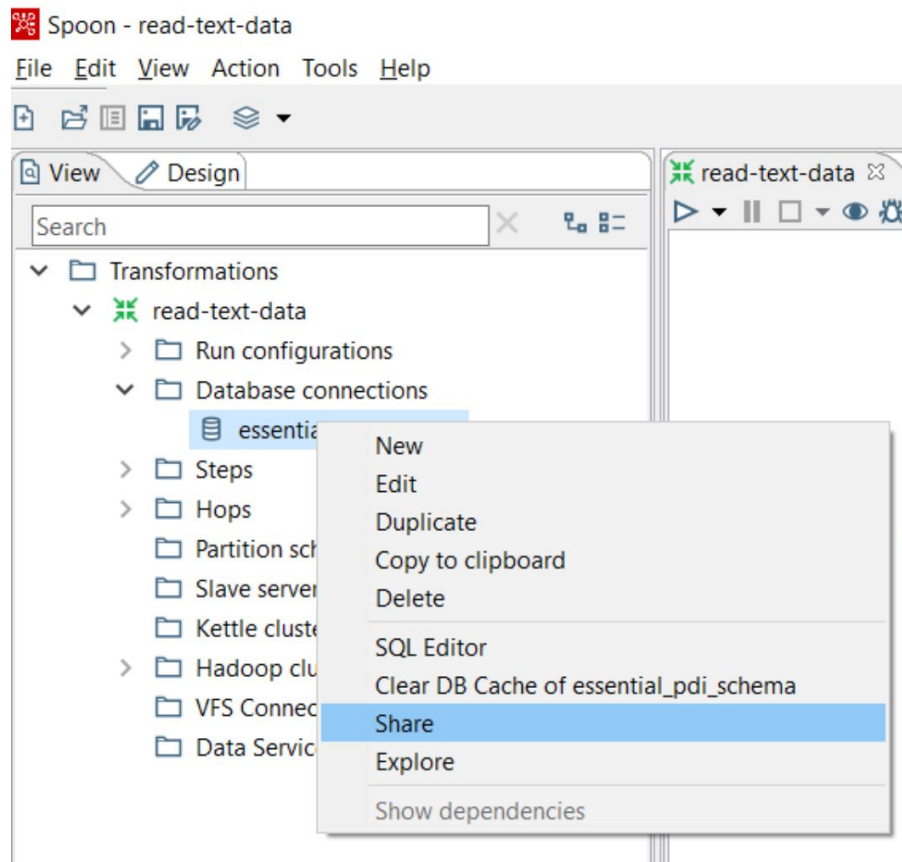
Whichever the connection method you choose, a setting window like the one you saw in the step-7 shows up allowing you to define the connection details.

Reuse the Existing Database Connection

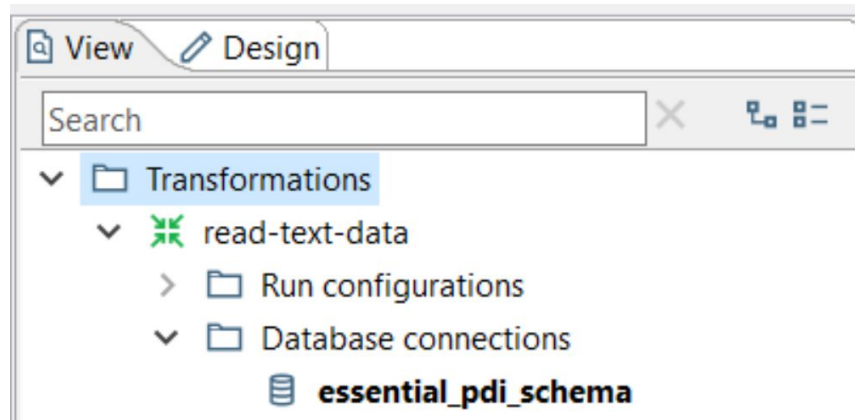
Most of the time when working in the real-world project, you wish to reuse the existing database connection in more than one transformation and jobs. In such cases it's recommended that you share the database connections.

Steps:

1. Launch Spoon.bat on Windows or Spoon.sh on Linux machine.
2. Open the existing transformation where the database connection is created.
3. Under the **View -> Transformation -> Database connections** section, right-click the database connection and click on **Share** as shown the screenshot:



Shared database connections are recognized in bold as shown in the screenshot:



Here the databases **essential_pdi_schema** is a shared connection. The shared connections are saved in a [shared.xml](#) in the Kettle folder user home

directory. Also, this shared database connection is saved in the transformation files.

Parameterizing the connection detail

Instead of hard-coding the connection values it is recommended to parameterize the connection values with the variables. Here we will edit the previous database connection we had created earlier and update the values with the variables.

Steps:

1. Launch Spoon.bat on Windows or Spoon.sh on Linux machine.
2. Open the existing transformation where the database connection is created.
3. Under the **View -> Transformation -> Database connections** section, right-click the database connection and click on **Edit** option.
4. Update the following variables in the database connection edit screen as shown in the screenshot:

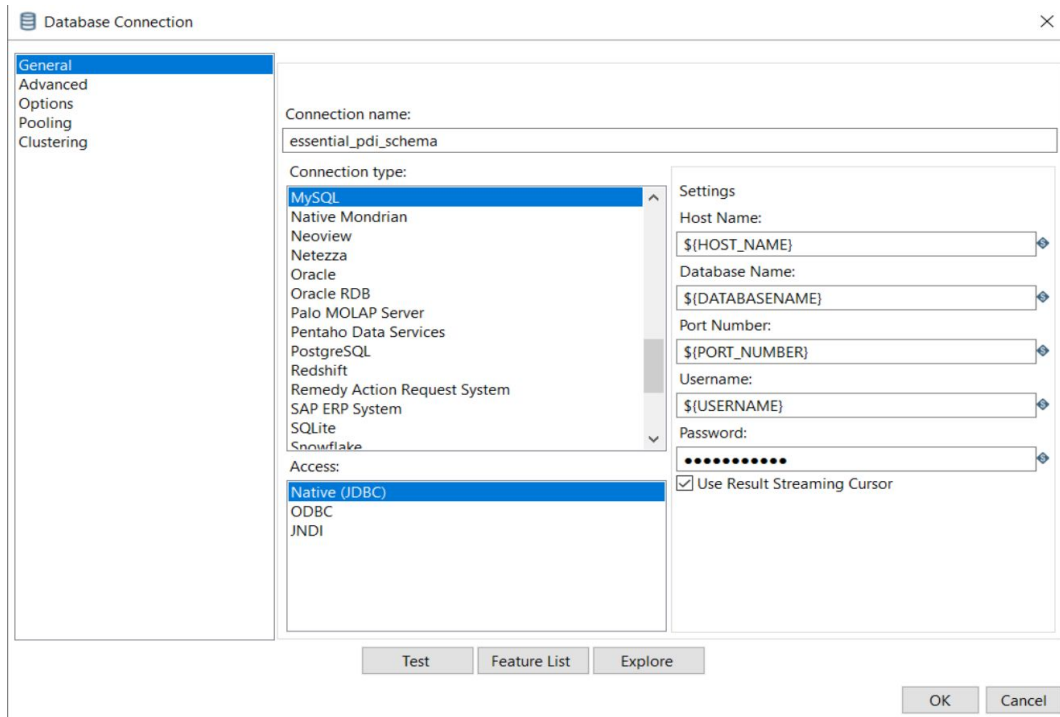
`${HOST_NAME}` – Hostname variable

`${DATABASENAME}` – Database name variable

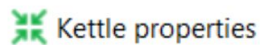
`${PORT_NUMBER}` – Port number variable

`${USERNAME}` – Username variable

`${PASSWORD}` – Password variable



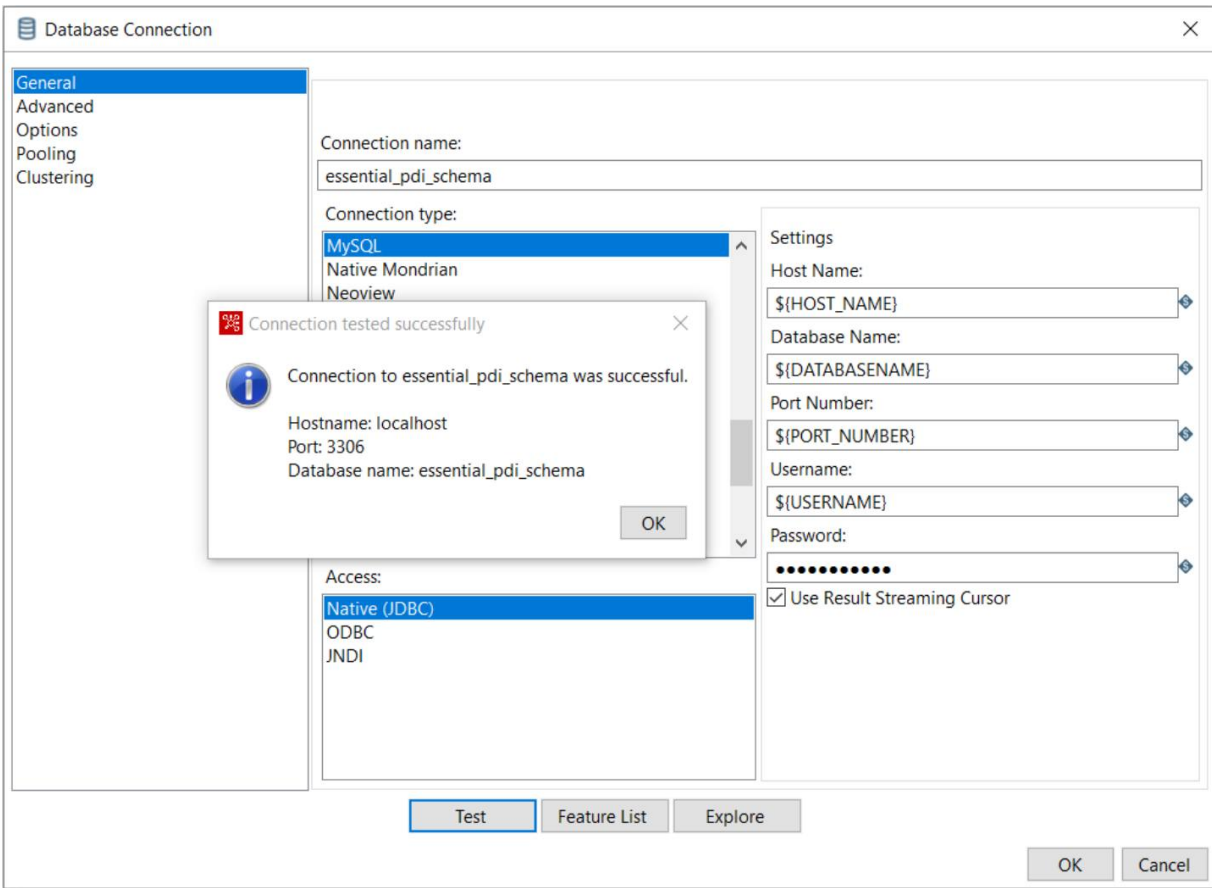
5. In the PDI Spoon, select Edit -> Edit the kettle.properties file option in the top-right corner of the screen.
6. The Kettle.properties window will appear.
7. Enter the variables and values as shown in screenshot:



Enter the values for the kettle.properties file

#	Variable name	Value
1	HOST_NAME	localhost
2	DATABASENAME	essential_pdi_schema
3	PORT_NUMBER	3306
4	USERNAME	root
5	PASSWORD	password

8. Now click **Test** button to test the database connection with variables settings.



This approach is very helpful if you decide to move the database from the local machine to a server and when you want move your jobs and transformations between different environments: development, QA, production. In such cases you just have to change the variable values in the kettle.properties file.

Getting data from files and writing data to a database table

Here we use the earlier transformations that we have created to read the txt files and modify the file to load the data into a target database table.

Reading TXT file and writing to a database table

ETL Flow:

Carry out the following steps:

1. Launch Spoon.bat on Windows or Spoon.sh on Linux machine.
2. Open read-text-data.ktr transformation file.
3. Drop a **Table Output** step (Output category) from the **Design** tab to the canvas area.
4. Create a hop connection as shown in the screenshot:



5. Select the connection details.
6. Enter the Target table information in the text area.
7. Check Specify database fields and click on **Get fields** button.
8. All the incoming fields is shown in the Fields to insert area as shown in the screenshot.

Table output

Step name: Table output

Connection: essential_pdi_schema [Edit...] [New...] [Wizard...]

Target schema: [Browse...]

Target table: customer_info [Browse...]

Commit size: 1000

Truncate table: ☐

Ignore insert errors: ☐

Specify database fields: ☒

Main options Database fields

Fields to insert:

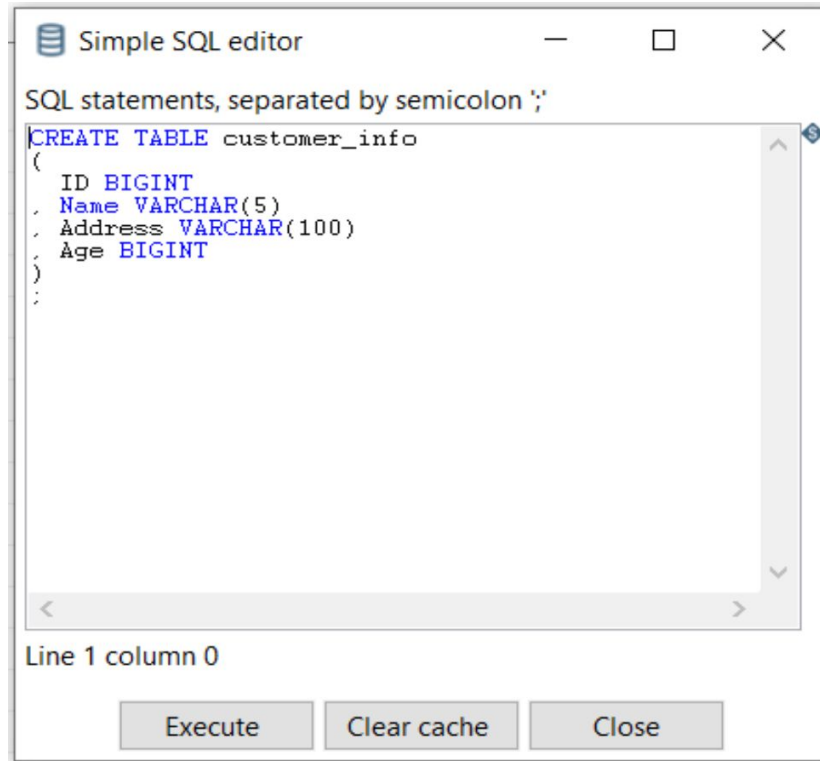
#	Table field	Stream field	
1	ID	ID	
2	Name	Name	
3	Address	Address	
4	Age	Age	

[Get fields]

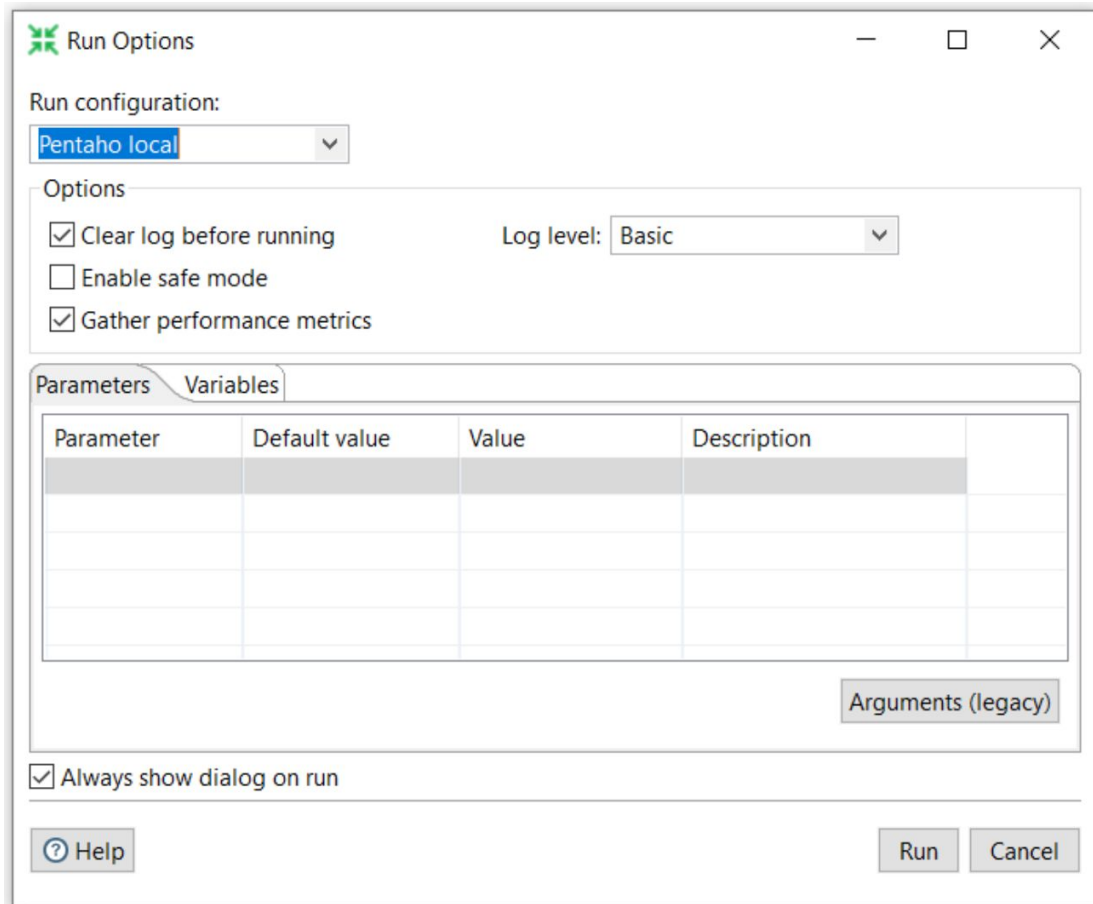
[Enter field mapping]

[Help] [OK] [Cancel] [SQL] ←

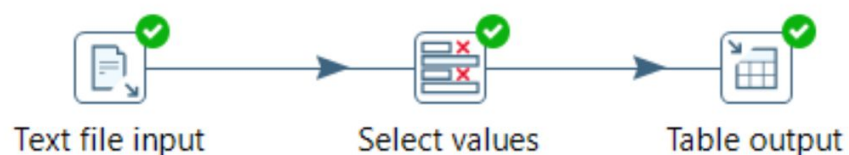
9. By clicking the **SQL** button, PDI Spoon will generate the DDL script to create a table as shown in the screenshot:



10. Click **Execute** and finally click the **OK** button.
11. With this, you have completed the ETL flow to load the txt data into a table. It's time for execution.
12. To Run the transformation, Go to **Action -> Run** (OR) click on **Play** button on top of the screen.
13. The **Run Options** window will appear prompting to select the Run configuration (Pentaho local) and Log levels (Basic).



14. The Log level field has different log writing options such as Error, Detailed, Minimal, Basic, Debug, Row Level (very detailed). Choose the options as per your requirements.
15. Click the **Run** button to execute the transformation file.
16. After successful execution, we should be able to see green tick icon on top of each steps in the transformation as shown in the screenshot:



17. The execution logging is shown in **Logging** tab under **Execution Results** section.

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2020/12/05 13:31:39 - RepositoriesMeta - Reading repositories XML file: C:\Users\Admin\kettle\repositories.xml
2020/12/05 13:31:45 - RepositoriesMeta - Reading repositories XML file: C:\Users\Admin\kettle\repositories.xml
2020/12/05 13:31:47 - Carte - Installing timer to purge stale objects after 1440 minutes.
2020/12/05 15:46:07 - Spoon - Running transformation using the Kettle execution engine
2020/12/05 15:46:07 - Spoon - Transformation opened.
2020/12/05 15:46:07 - Spoon - Launching transformation [read-text-data]...
2020/12/05 15:46:07 - Spoon - Started the transformation execution.
2020/12/05 15:46:08 - read-text-data - Dispatching started for transformation [read-text-data]
2020/12/05 15:46:08 - Table output.0 - Connected to database [essential_pdi_schema] (commit=1000)
2020/12/05 15:46:08 - Text file input.0 - Opening file: file:///C:/essential_pentaho_etl/pdi-source-code/source-files/source-customer-info.txt
2020/12/05 15:46:08 - Text file input.0 - Finished processing (I=5, O=0, R=0, W=4, U=1, E=0)
2020/12/05 15:46:08 - Select values.0 - Finished processing (I=0, O=0, R=4, W=4, U=0, E=0)
2020/12/05 15:46:08 - Table output.0 - Finished processing (I=0, O=4, R=4, W=4, U=0, E=0)
2020/12/05 15:46:08 - Spoon - The transformation has finished!!

18. The step operations are shown in **Step Metrics** tab under **Execution Results** section.

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

#	Stepname	Copypnr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	Text file input	0	0	4	5	0	1	0	0	Finished	0.1s	58	-
2	Select values	0	4	4	0	0	0	0	0	Finished	0.1s	44	-
3	Table output	0	4	4	0	4	0	0	0	Finished	0.3s	14	-

19. The output data is shown in **Preview data** tab under **Execution Results** section.

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

☒ First rows ☐ Last rows ☐ Off

#	ID	Name	Address	Age
1	1101	Jack	Newyork	20
2	1102	Ilisa	Australia	25
3	1103	Kumar	Mysore	23
4	1104	Kavan	Bangalore	26

Exercise Time!

It's time for you to do some quick crunching. Use the previous transformations that we have created to read the csv and xml files and modify the transformation file to load the data into a target database table.

Here is the reference section for the previous transformation file:

1. Reading CSV file and writing to a database table
2. Reading XML file and writing to a database table

Getting data from a database tables and writing data to a database table

In the Pentaho BI/DWH project, we use PDI Spoon for the ETL process. Extracting data from source databases for transforming, loading into the Data Warehouse/Datamart databases for generating business reports, dashboards, Adhoc report analysis and so on. Using Spoon GUI, we will learn the simplest way to do the database reading and writing.

Read and Write Database table

ETL Flow:

Carry out the following steps:

1. Launch Spoon.bat on Windows or Spoon.sh on Linux machine.
2. Create a new transformation.
3. Drop a **Table input** step into the canvas.

4. From the **Connection** drop-down list, select the connection to the database where your data resides, or create it if it doesn't exist.
5. In SQL text area, enter the following SQL statement:
`SELECT * FROM essential_pdi_schema.customer_info;`
6. From the **Flow** category, add a **Filter rows** step and filter the result set on *age* ≥ 25 condition as shown in the screenshot:

Filter rows

Step name: Filter rows

Send 'true' data to step: Table output

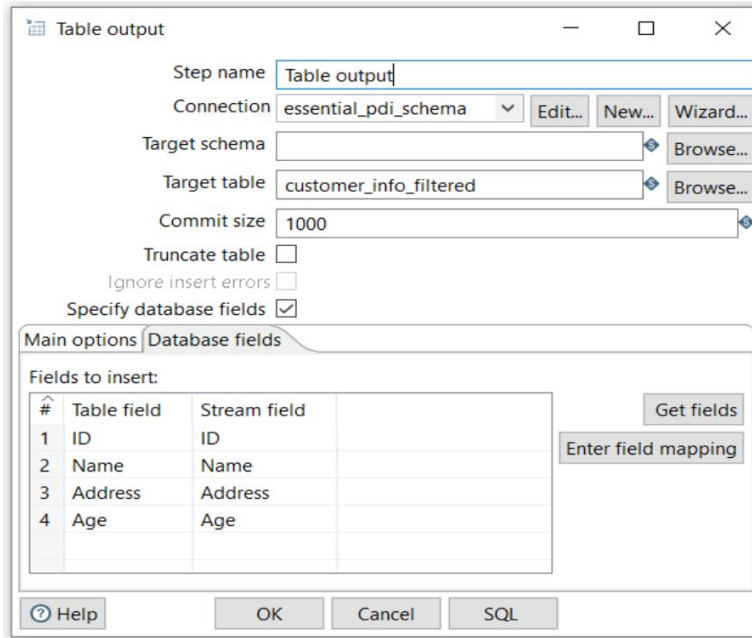
Send 'false' data to step:

The condition:

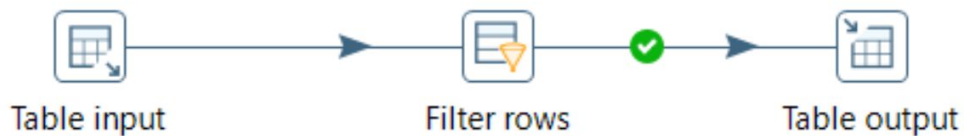
Age >= 25 (Integer)

Help OK Cancel

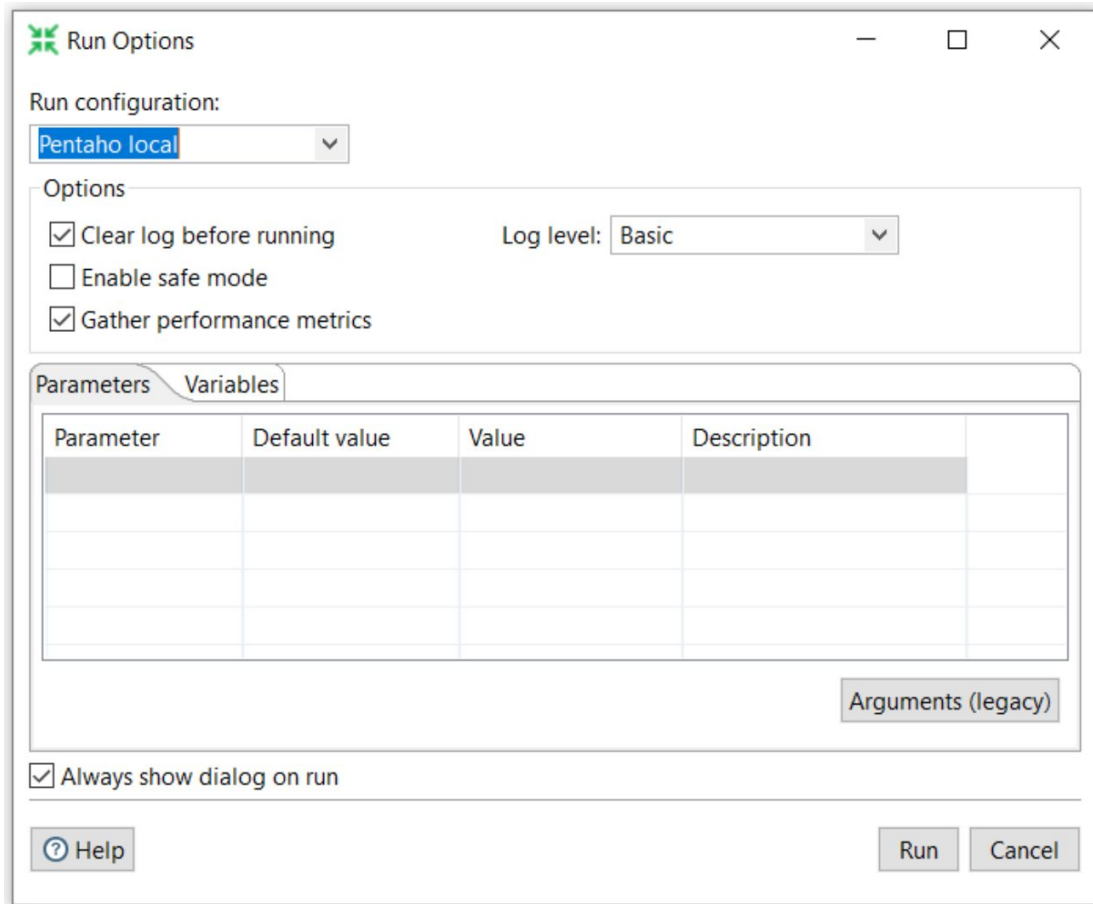
7. From the **Output** category, add a **Table output** step and select incoming fields, database connections, enter the table name as shown in the screenshot:



8. Generate and execute the DDL statement by clicking on **SQL** and **Execute** button respectively.
9. Now you should be able to see the database table created in the back-end database.
10. The hop connection between the steps is shown in the screenshot:



11. With this, you have completed the ETL flow to read and load the data from and to the table respectively.
12. To Run the transformation, go to Action -> Run (OR) click on Play button on top of the screen.
13. The Run Options window will appear prompting to select the Run configuration (Pentaho local) and Log levels (Basic).



The image shows a 'Run Options' dialog box with a green icon in the top-left corner. The title bar contains the text 'Run Options' and standard window controls (minimize, maximize, close). The main area is divided into several sections:

- Run configuration:** A dropdown menu showing 'Pentaho local'.
- Options:** A group box containing:
 - ☒ Clear log before running
 - ☐ Enable safe mode
 - ☒ Gather performance metrics
 - Log level:** A dropdown menu showing 'Basic'.
- Parameters / Variables:** A tabbed interface with 'Parameters' selected. It contains a table with the following structure:

Parameter	Default value	Value	Description
- Arguments (legacy):** A text area for entering legacy arguments.
- Always show dialog on run:** A checkbox that is checked.
- Buttons:** 'Help' (with a question mark icon), 'Run', and 'Cancel' are located at the bottom.

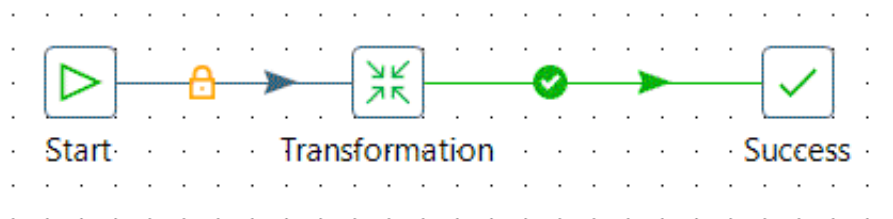
14. The Log level field has different log writing options such as Error, Detailed, Minimal, Basic, Debug, Row Level (very detailed). Choose the options as per your requirements.
15. Click the **Run** button to execute the transformation file.
16. After successful execution, a new data will be loaded into the target table containing the list of customer info whose *age* ≥ 25 .

Reading data from a database by providing Table input parameters.

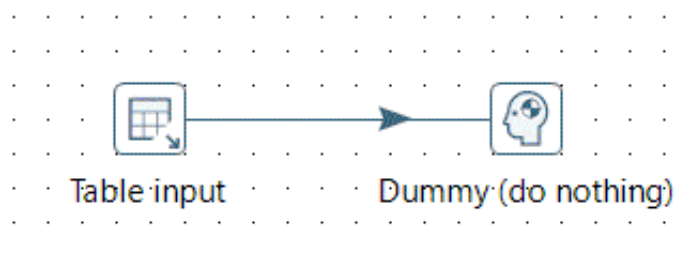
ETL Flow:

Carry out the following steps:

1. Launch Spoon.bat on Windows or Spoon.sh on Linux machine.
2. Create a new Job and save the file as “ read-parameter-input.kjb ”.
3. Drop **Start**, **Transformation** and **Success** entries from **General** design category and create hops as shown in the screenshot.



4. Create a new Transformation and save the file as “ read-parameterized-table-input.ktr ”
5. Drop a **Table input** step into the canvas area.
6. From the **Connection** drop-down list, select the connection to the database where your data resides, or create it if it doesn't exist.
7. In SQL text area, enter the following SQL statement:
`SELECT * FROM essential_pdi_schema.customer_info WHERE ID=${ID};`
8. Tick **Replace variables in script** checkbox and click Ok button.
9. Click **CTRL+T** a Transformation properties window will appear. Go to Parameters tab and add **ID** parameter and set **1101** as a default value and click **OK** button.
10. Drop the **Dummy (do nothing)** step (Flow Category) into the canvas area.
11. Create the hop connection as shown in the screenshot:



12. Now open “read-parameterized-input.kjb” job file and edit the **Transformation** step and select the “read-parameterized-table-input.ktr” file.
13. With this, you have completed the ETL flow to read the data from table using the parameterized SQL query.
14. To Run the Job, go to **Action -> Run** (OR) click on **Play** button on top of the screen.
15. The Run Options window will appear prompting to select the Run configuration (Pentaho local) and Log levels (Basic).

Run Options

Run configuration:
Pentaho local

Options

☐ Expand remote job Log level: Debug

☒ Clear log before running Start job at:

☐ Enable safe mode

☐ Gather performance metrics

Parameters Variables

Parameter	Default value	Value	Description

Arguments (legacy)

☒ Always show dialog on run

Help Run Cancel

16. The Log level field has different log writing options such as Error, Detailed, Minimal, Basic, Debug, Row Level (very detailed). Choose the options as per your requirements.
17. Click the Run button to execute the ETL Job.

18. After successful execution, you are only reading the parameterized table input data.

Exercise Time!

It's time for you to do some quick crunching.

Create a new ETL work flow for;

1. Reading multiple csv files in a single ETL transformation.
2. Writing file data into multiple target types i.e., table output and text output.
3. Reading and parsing the semi-structured data.
4. Inserting data into a table.
5. Updating data in a table.
6. Deleting data in a table.

Working with Rest API's

In some of BI/DWH projects, you will be asked to read the data from the source servers using REST API's. Pentaho PDI Spoon offers an easy step to work with REST API's. Here let us understand the REST API's Get and Post methods.

GET Method

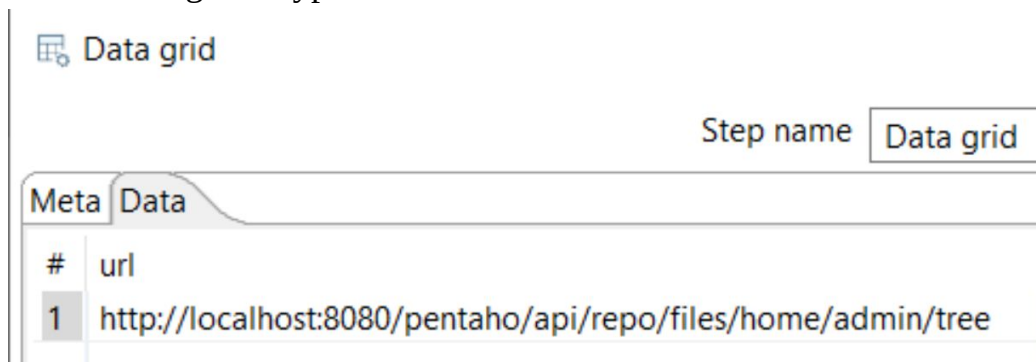
GET METHOD

In this sample, we are reading and parsing the REST API contents inside **/home/admin** folder path.

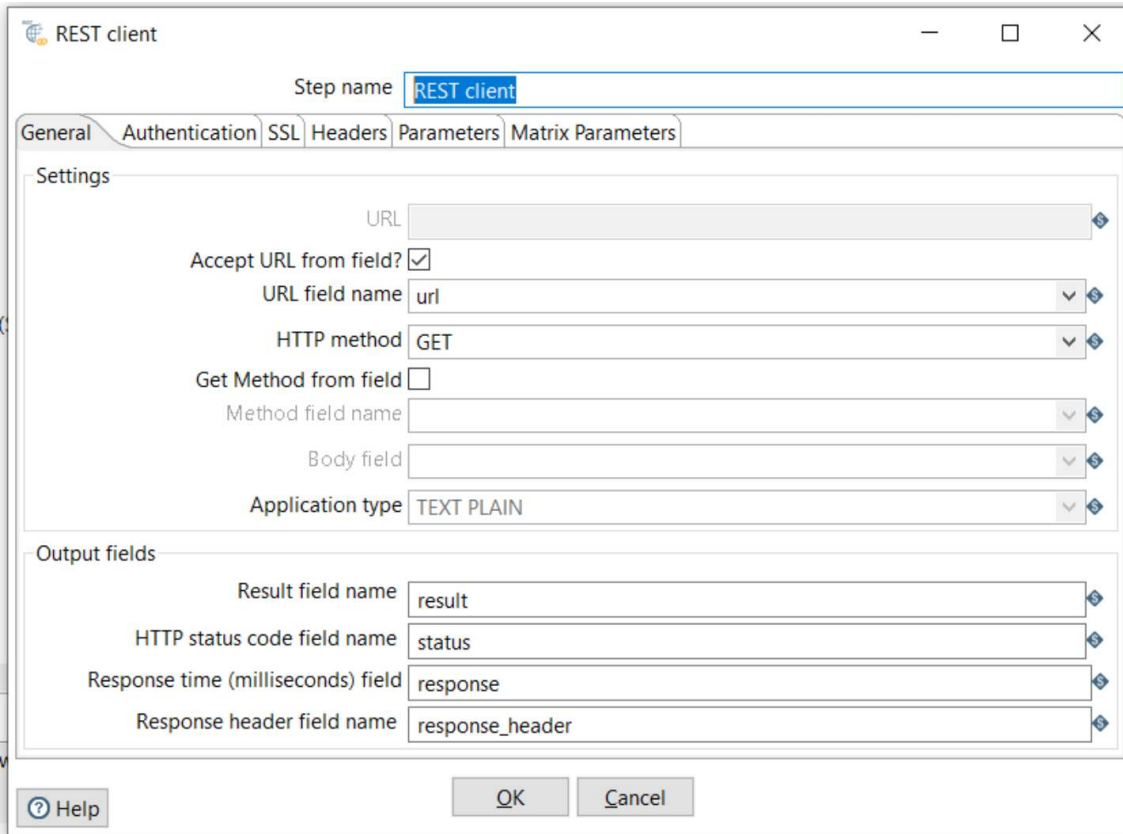
ETL Flow:

Carry out the following steps:

1. Launch Spoon.bat on Windows or Spoon.sh on Linux machine.
2. Create a new transformation.
3. Drop a **Data Grid** step into the canvas area and enter a new field 'URL' with String data type and insert value as shown in the screenshot:



4. Drop a **REST client** step into the canvas area and enter the details as shown in the screenshot:



Check the Accept URL from field option and choose the *URL* field in the **URL Field name** dropdown.

Choose Get option in the **HTTP method** dropdown.

Enter the field name for **Result field name**

Enter the field name for **HTTP status code field name**

Enter the field name for **Response time (milliseconds) field name**

Enter the field name for **Response header field name**

5. Drop a **Select values** step into the canvas area and select all the incoming fields. In the Meta-data tab, select the **URL** field and change the datatype to string and click **Ok** button.
6. Drop an **XML input stream (StAX)** step into the canvas area and choose the URL field as shown in the screenshot:

XML input stream (StAX)

Step name: XML input stream (StAX)

Filename:

Source is from a previous step: ☒

Source field name: result

Add filename to result?: ☐

Skip (Elements/Attributes): 0

Limit (Elements/Attributes): 0

Default String Length (Elements/Attributes): 1024

Encoding: UTF-8

Add Namespace information?: ☐

Trim strings?: ☒

Include filename in output?: ☐ Fieldname: xml_filename

Row number in output?: ☐ Fieldname: xml_row_number

XML data type (numeric) in output?: ☐ Fieldname: xml_data_type_numeric

XML data type (description) in output?: ☒ Fieldname: xml_data_type_description

XML location line in output?: ☐ Fieldname: xml_location_line

XML location column in output?: ☐ Fieldname: xml_location_column

XML element ID in output?: ☒ Fieldname: xml_element_id

XML parent element ID in output?: ☒ Fieldname: xml_parent_element_id

XML element level in output?: ☒ Fieldname: xml_element_level

XML path in output?: ☒ Fieldname: xml_path

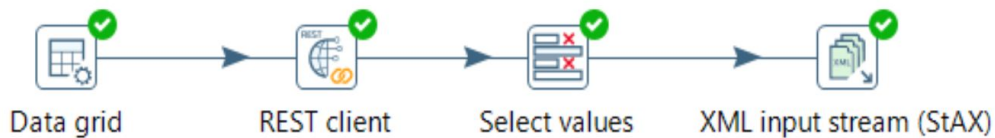
XML parent path in output?: ☒ Fieldname: xml_parent_path

XML data name in output?: ☒ Fieldname: xml_data_name

XML data value in output?: ☒ Fieldname: xml_data_value

Help OK Preview Cancel

7. Transformation flow should look as shown in the screenshot:



8. With this, you have completed the ETL flow to read the REST API data using Get method. It's time for execution.

9. To Run the transformation, go to **Action -> Run** (OR) click on **Play** button on top of the screen.

10. The **Run Options** window will appear prompting to select the Run configuration (Pentaho local) and Log levels (Basic).

Run Options

Run configuration:
Pentaho local

Options

☒ Clear log before running Log level: Basic

☐ Enable safe mode

☒ Gather performance metrics

Parameters Variables

Parameter	Default value	Value	Description

Arguments (legacy)

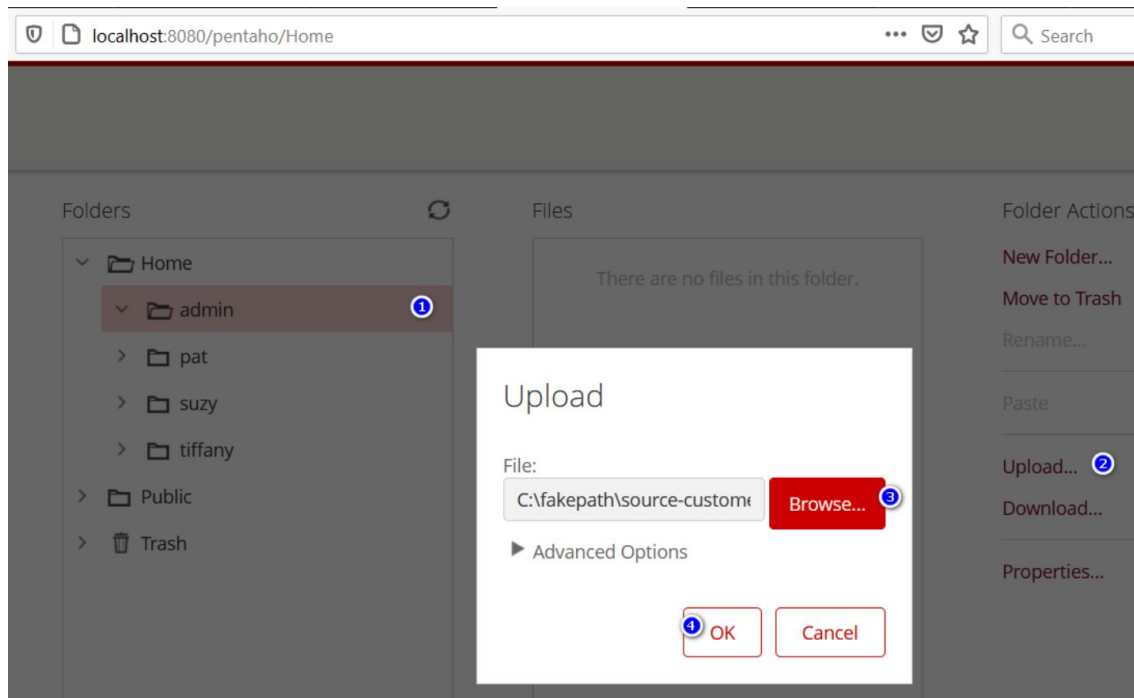
☒ Always show dialog on run

Help Run Cancel

11. The Log level field has different log writing options such as Error, Detailed, Minimal, Basic, Debug, Row Level (very detailed). Choose the options as per your requirements.
12. Click the Run button for executing the transformation.
13. After successful execution, REST API data is read and processed within the PDI spoon.

PUT Method

For this example, let us upload a sample source-product-info.xml file to the Pentaho BI Server Home -> Admin folder. Later we will delete this file using the REST API calls.



ETL Flow:

Carry out the following steps:

1. Create a new transformation.
2. Copy all the steps discussed in the Get Method sections.
3. In the **Filter rows & Filter rows 2** steps, the custom requirement filters are applied.
4. In the **Select values 2** step, we are removing unnecessary fields.
5. In the **Sort rows** step, we are sorting the key field.
6. In the **Row demoralizer** step, we deformalizing the result set as shown in the screenshot:

Row denormaliser

Step name

The key field

The fields that make up the grouping:

#	Group field
1	xml_parent_element_id

Target fields:

#	Target fieldname	Value fieldname	Key value	Type
1	id	xml_data_value	id	String
2	name	xml_data_value	name	String

< >

7. In the **Filter rows 3** step, we are ignoring records as per the requirement.
8. Go to **Edit -> Edit the Kettle.properties file** option and create the following 2 variables.

Variable Name	Value
REST_GET_CALL	http://localhost:8080/pentaho/api/repo/files/home/admin/tree
REST_PUT_CALL	http://localhost:8080/pentaho/api/repo/files/deletepermanent

9. Drop the **REST client** step and update the fields as shown in the screenshot:

REST client

Step name: **REST client 2**

General | Authentication | SSL | Headers | Parameters | Matrix Parameters

Settings

URL: **\${REST_PUT_CALL}**

Accept URL from field? ☐

URL field name: **url_delete**

HTTP method: **PUT**

Get Method from field? ☐

Method field name:

Body field: **id**

Application type: **OCTET STREAM**

Output fields

Result field name: **result**

HTTP status code field name: **status**

Response time (milliseconds) field: **response**

Response header field name: **response_header**

Help OK Cancel

Here we are updating REST Client (PUT method) request details.

URL - Enter the PUT variable name

HTTP Method - PUT

Body Field - id field

Application type - OCTET STREAM

Enter the field name for **Result field name**

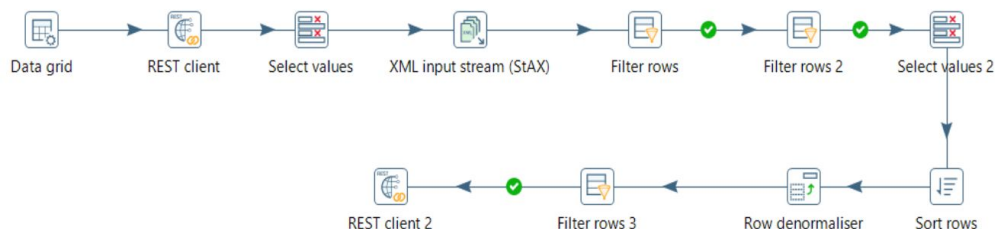
Enter the field name for **HTTP status code field name**

Enter the field name for **Response time (milliseconds) field name**

Enter the field name for **Response header field name**

10. Edit the REST client step (GET Method) and update URL text area with `${REST_GET_CALL}` variable.

11. With this, you have completed the ETL flow to read the REST API data using Get method and delete the server files using REST PUT method.



12. It's time for execution. To Run the transformation, go to **Action -> Run** (OR) click on **Play** button on top of the screen.
13. The Run Options window will appear prompting to select the Run configuration (Pentaho local) and Log levels (Basic).

Run Options

Run configuration:
Pentaho local

Options

☒ Clear log before running Log level: Basic

☐ Enable safe mode

☒ Gather performance metrics

Parameters Variables

Parameter	Default value	Value	Description

Arguments (legacy)

☒ Always show dialog on run

Help Run Cancel

14. The Log level field has different log writing options such as Error, Detailed, Minimal, Basic, Debug, Row Level (very detailed). Choose the options as per your requirements.
15. Click the **Run** button for executing the transformation.
16. After successful execution, the source-product-info.xml file is read and later it is deleted using GET and PUT API calls respectively.

Exercise Time!

It's time for you to do some quick crunching. Create a new ETL work flow to;

1. Using `/repo/files/{pathId }/rename` REST API call, rename a file in the Pentaho PUC.
2. Using `/repo/files/{pathId }/properties` REST API call,

Action Plan

Take up all the exercises mentioned in this chapter. You are always free to explore other PDI design steps.

Chapter 5: Scheduling and Managing

In this chapter you'll learn about the following:

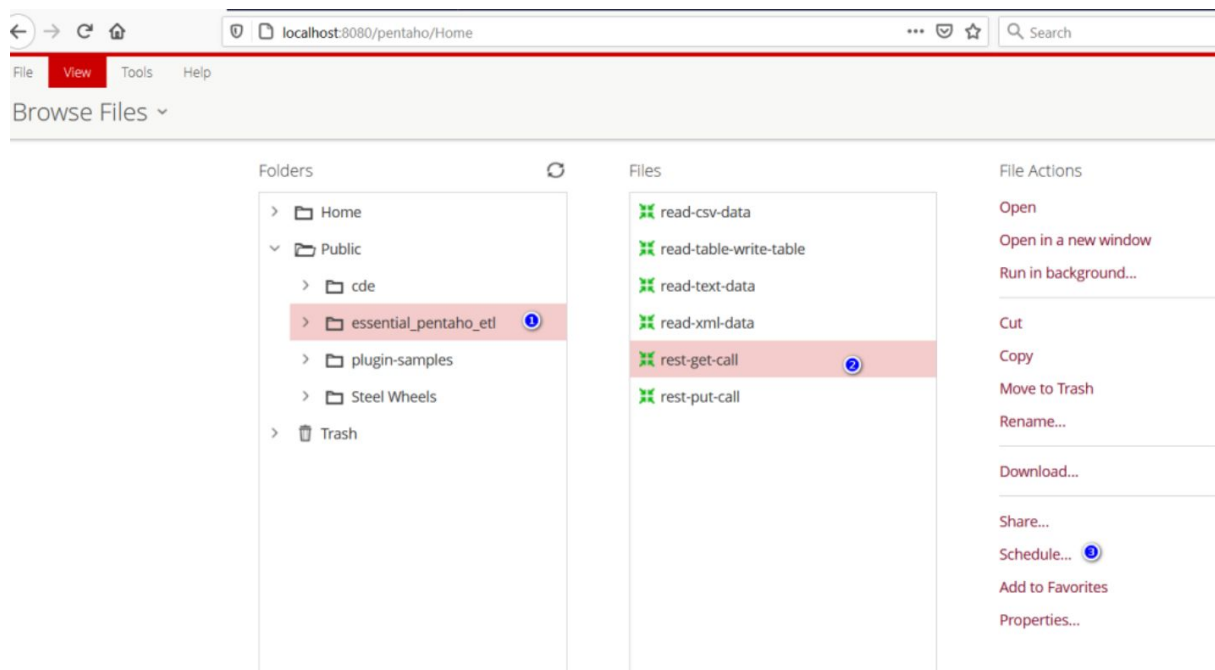
- How to schedule the ETL Jobs and Transformation on PDI server.
- How to monitor the scheduled ETL Jobs and Transformation on PDI server.

ETL Scheduling

In the BI/DWH projects, data has to be populated on a daily basis in the production environment for the reporting solution. You design the ETL data flow and workflow using PDI Spoon do achieve this work. In order to automate the data extraction, transformation and loading process, you schedule these Jobs and Transformations to run on the Pentaho server. The scheduled ETL's runs at different intervals and at different levels to load the data from source to staging to data warehouse and data marts in the production environments. Using Pentaho scheduler feature it is easy to schedule the ETL jobs and Transformations. Let us understand how to achieve this.

STEPS

1. In the Pentaho server, select the KTR/KJB file that you wish to schedule and click on **Schedule** option in the **File Actions** section in the right side of the screen.



2. A New Schedule window appears asking to enter the **Schedule name** and select Generated **content location** path. Enter the details as shown in the screenshot and click **Next** button.

New Schedule

Schedule Name: *This will also be the name of the generated content.*

☐ Append timestamp to generated content

Generated Content Location:

☐ Overwrite existing files with same name

3. Enter the *Schedule date & time* and *recurrences* and click **Finish** button.

New Schedule

Recurrence:

Run Once ▼

Start Time

09 ▼ 37 ▼ PM ▼ India Daylight Time (UTC+0530) ▼

Start Date

12/5/20

« < Dec 2020 > »

S	M	T	W	T	F	S
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9

Back Finish Cancel

4. A new schedule is created and server will prompt you to switch to the Schedule Tab. Click **Yes** button.

Schedule Created

Your new schedule was successfully created and is listed in the Schedules perspective. Would you like to view your list of schedules?

Yes No

5. In the Schedule tab, you can see the new schedule is listed with the specified parameters.

File View Tools Help

Schedules ▾ admin ▾

Manage Schedules

Schedule Name ▾	Repeats	Source File	Output Location	Last Run	Next Run	Created By	Status
PentahoSystemVersionCheck	Every day at 7:02:56 PM	PentahoSystemVersionCheck	-	Dec 5, 2020 7:02:56 PM	Dec 6, 2020 7:02:56 PM	admin	Normal
rest-get-call	Run Once	/public/essential_pentaho_etl/rest-get-call	/home/admin	-	Dec 5, 2020 9:37:00 PM	admin	Normal

Blockout Times

No blockout times for scheduling have been created.

Create Blockout Time

Use the same approach to schedule the PDI Jobs.

ETL Monitoring

In BI/DWH projects, once the ETL's are moved to the production, there is a need for daily monitoring of ETL Jobs and Transformations. In order to effectively monitor these data flows and work flows, Pentaho offers the web-based console page called **Kettle Status** web page. Using the Kettle Status web page, you can remotely monitor the status of all the scheduled Jobs and Transformations running on the Pentaho server. The admin and maintenance team don't have to login to the Remote machine to see the statuses of the Job and transformations. This is very helpful in variety of reasons. The admin and support engineers can review the logs of the Transformation and Jobs within the Kettle Status web page. Let us see how to achieve this in action.

STEPS

1. Make sure the Pentaho server is running.
2. Schedule or Run any transformations on the server as shown in the previous ETL Scheduling section.
3. After scheduling or executing the job on the server, go to <http://localhost:8080/pentaho/kettle/status> URL as shown in the

screenshot.

The screenshot shows the Pentaho Kettle Status web interface. The browser address bar indicates the URL is `localhost:8080/pentaho/kettle/status`. The page title is "PDI Status".

Transformations

Below the title is a table with the following data:

Name	Carte Object ID	Status	Last log date	Last log time
rest-get-call	69bc3a2b-8c7b-40eb-9835-46adf47bb96e	Finished	2020/12/05	21:37:00.481

Below the table are icons for play, square, eye, and close.

Jobs

Below the title is a table with the following data:

Name	Carte Object ID	Status	Last log date	Last log time
------	-----------------	--------	---------------	---------------

Below the table are icons for play, square, eye, and close.

Configuration details:

The maximum size of the central log buffer 10000 lines
The maximum age of a log line 2880 minutes
The maximum age of a stale object 240 minutes
Repository name singleDISEverInstance

These parameters can be set in the slave server configuration XML file: C:\essential_pentaho_etl\pentaho-server-ce-9.1.0.0-324\pentaho-server\pentaho-solutions\system\kettle\slave-server-config.xml

4. In the Kettle status screen, you will see the list of KTR/KJB execution details.
5. In order to view the detail information of each Job or Transformation execution, click on the respective file and click on the **eye** icon.
6. The detailed status information of the Job or Transformation is shown in the screenshot:

rest-get-call status

[← Back to Carte status](#)



Carte Object ID	Status	Last log date	
69bc3a2b-8c7b-40eb-9835-46adf47bb96e	Finished	2020/12/05	View as XML

Step detail

Step name	CopyNr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed	pr/in/out
REST client	0	1	1	0	0	0	0	0	Finished	0.4	2	-
Select values	0	1	1	0	0	0	0	0	Finished	0.4	2	-
XML input stream (StAX)	0	1	151	151	0	0	0	0	Finished	0.4	343	-
Data grid	0	0	1	0	0	0	0	0	Finished	0.0	28	-

In order to access the Kettle status page from remote machine, access the URL with the IP address or hostname of the machine.

Action Plan

Schedule any sample Jobs and Transformations on the Pentaho server and later monitor the status using Pentaho kettle status.

SUMMARY

This book is divided into 5 chapters, summarized here:

Chapter 1: Getting Started with PDI — Chapter 1 relates to understand the working environment of PDI server, PDI client tool Spoon/KETTLE. Understand the installation steps for Pentaho server and Pentaho Data Integration (PDI) Spoon.

Chapter 2: Customizing PDI — Chapter 2 relates customizing the PDI Spoon client tool and understand the PDI marketplace.

Chapter 3: Configuring PDI — Chapter 3 relates to configuration of the PDI Spoon tool. This chapter helps to understand the system configuration for PDI server, understand the system configuration for PDI client, understand the components references, how to install Pentaho Data Integration (PDI) server, how to install Pentaho Data Integration (PDI)/Spoon/Kettle client and understand how to install database drivers.

Chapter 4: Working with PDI — Chapter 4 relates to dealing with different types of data. Reading and writing files, reading and writing database tables, reading REST API data- Get & POST method requests.

Chapter 5: Scheduling and Monitoring— Chapter 5 relates to scheduling and monitoring the ETL jobs and transformations on the PDI server.

SAMPLE CODE USED IN THIS BOOK

This book provides a handful of sample code, most of it in the form of complete ETL Job/Transformation programs. Thus, the examples were tested on a Windows 10 system.

Those ETLs using PDI 9 features require Java supporting those features, but the most of ETLs should work with Pentaho version 6 and above on a compliant system.

The sample code for the complete ETLs described in this book is available on the Author's website. Refer the link given in the About Author section for more information.

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